2.18 Trigonometry: Harmonic Forms

Worked Solutions

1 a)

$$R = \sqrt{8^2 + 15^2} = 17$$

1 b)

$$R = \sqrt{12^2 + 35^2} = 37$$

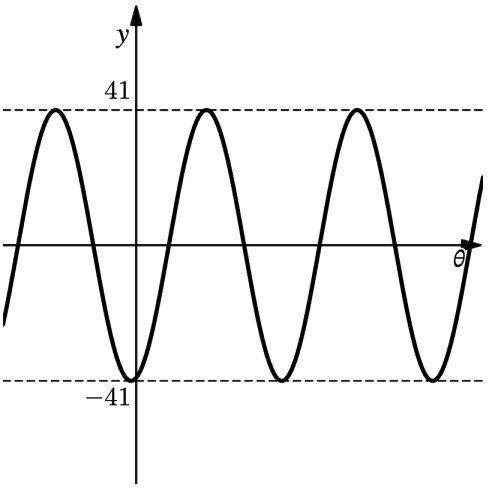
1 c)

$$R = \sqrt{14^2 + 20^2} = 2\sqrt{149}$$

1 d)

$$R = \sqrt{(2\sqrt{2})^2 + (5\sqrt{2})^2} = \sqrt{58}$$

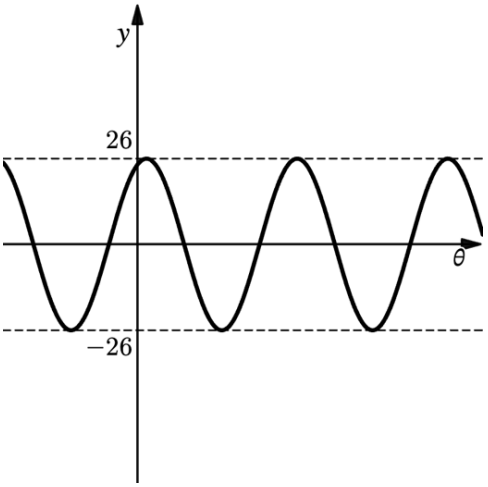
2 a)



$$R = \sqrt{9^2 + 40^2} = 41$$

Max = 41, Min = −41.

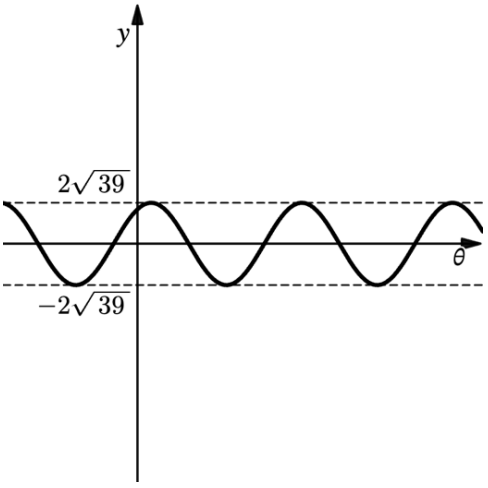
2 b)



$$R = \sqrt{10^2 + 24^2} = 26$$

Max $= 26$, Min $= -26$.

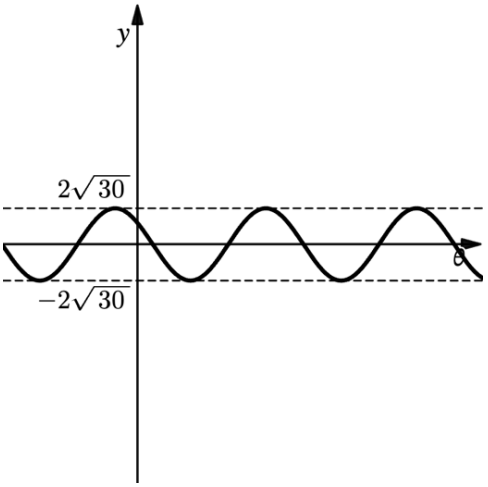
2 c)



$$R = \sqrt{(6\sqrt{3})^2 + (4\sqrt{3})^2} = 2\sqrt{39}$$

Max $= 2\sqrt{39}$, Min $= -2\sqrt{39}$.

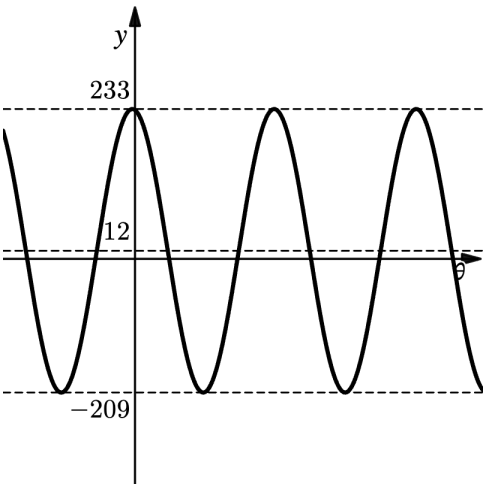
2 d)



$$R = \sqrt{(3\sqrt{5})^2 + (5\sqrt{3})^2} = 2\sqrt{30}$$

Max $= 2\sqrt{30}$, Min $= -2\sqrt{30}$.

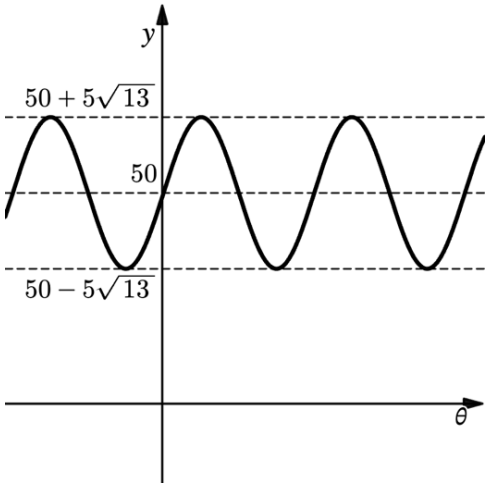
3 a)



$$R = \sqrt{220^2 + 21^2} = 221$$

Max $= 12 + 221 = 233$, Min $= 12 - 221 = -209$.

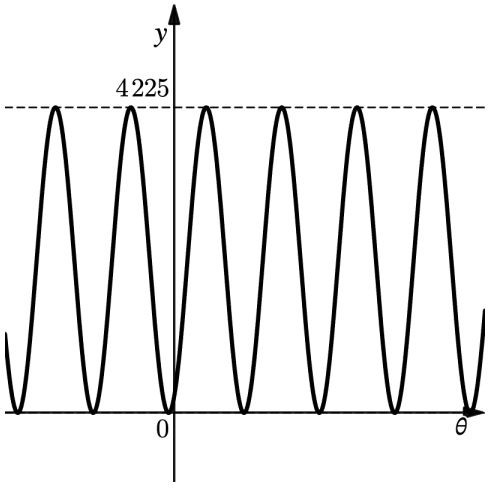
3 b)



$$R = \sqrt{1^2 + 18^2} = 5\sqrt{13}$$

Max $= 50 + 5\sqrt{13}$, Min $= 50 - 5\sqrt{13}$.

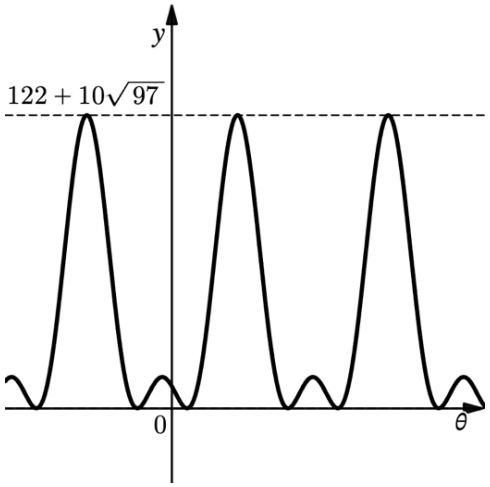
3 c)



$$R = \sqrt{16^2 + 63^2} = 65$$

Max $= 65^2 = 4225$, Min $= 0$.

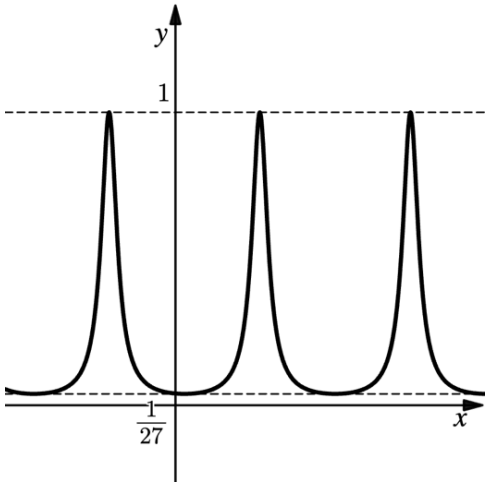
3 d)



$$R = \sqrt{4^2 + 9^2} = \sqrt{97}$$

Max $= (5+\sqrt{97})^2 = 122 + 10\sqrt{97}$, Min $= 0$.

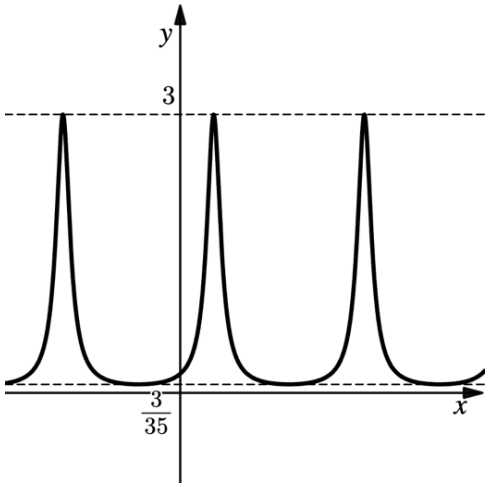
4 a)



$$R = \sqrt{5^2 + 12^2} = 13$$

Max $= \frac{1}{14-13} = 1$, Min $= \frac{1}{14+13} = \frac{1}{27}$.

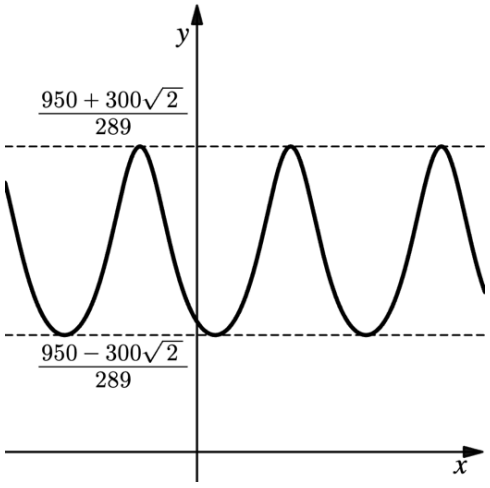
4 b)



$$R = \sqrt{84^2 + 13^2} = 85$$

$\text{Max} = \frac{15}{90-85} = 3$, $\text{Min} = \frac{15}{90+85} = \frac{3}{35}$.

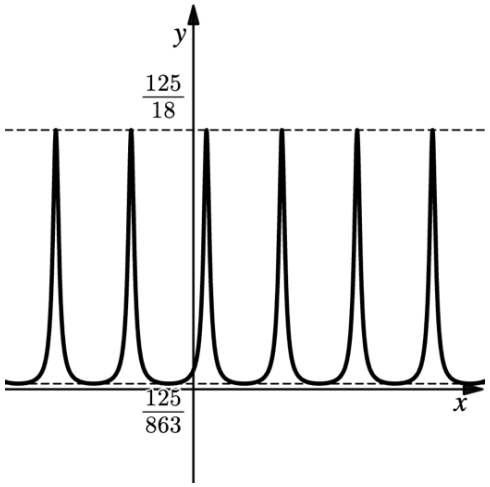
4 c)



$$R = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\text{Max} = \frac{100}{(6-\sqrt{2})^2} = \frac{950+300\sqrt{2}}{289}, \quad \text{Min} = \frac{100}{(6+\sqrt{2})^2} = \frac{950-300\sqrt{2}}{289}.$$

4 d)



$$R = \sqrt{33^2 + 56^2} = 65$$

$\text{Max} = \frac{625}{90+0} = \frac{125}{18}$, $\text{Min} = \frac{625}{90+65^2} = \frac{125}{863}$.

5)

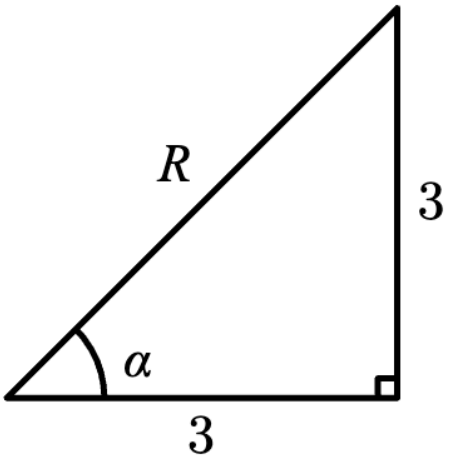
$$\text{Max} = \sqrt{10 + \sqrt{(3\sqrt{3})^2 + k^2}} = 4$$

$$\Rightarrow 10 + \sqrt{27 + k^2} = 16 \Rightarrow \sqrt{27 + k^2} = 6$$

$$\Rightarrow 27 + k^2 = 36 \Rightarrow k^2 = 9 \Rightarrow k = \pm 3$$

6 a)

$$R\sin(x+\alpha) \equiv R(\sin x\cos\alpha + \cos x\sin\alpha)$$
$$\equiv R\cos\alpha\sin x + R\sin\alpha\cos x$$
$$3\sin x + 3\cos x \equiv R\cos\alpha\sin x + R\sin\alpha\cos x$$
$$\therefore 3 = R\cos\alpha \text{ and } 3 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{3}{R} \text{ and } \sin\alpha = \frac{3}{R}$$



$$R = \sqrt{3^2+3^2} = 3\sqrt{2}$$
$$\alpha = \arctan\left(\frac{3}{3}\right) = 45^\circ$$
$$\therefore 3\sin x + 3\cos x = 3\sqrt{2}\sin(x+45^\circ)$$

6 b)

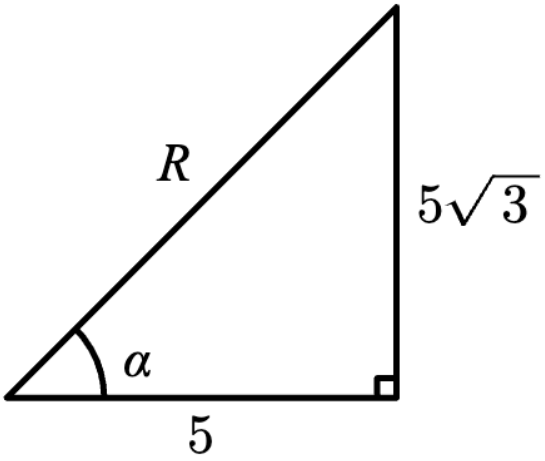
$$R\sin(x+\alpha) \equiv R(\sin x\cos\alpha + \cos x\sin\alpha)$$

$$\equiv R\cos\alpha\sin x + R\sin\alpha\cos x$$

$$5\sin x + 5\sqrt{3}\cos x \equiv R\cos\alpha\sin x + R\sin\alpha\cos x$$

$$\therefore 5 = R\cos\alpha \text{ and } 5\sqrt{3} = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{5}{R} \text{ and } \sin\alpha = \frac{5\sqrt{3}}{R}$$



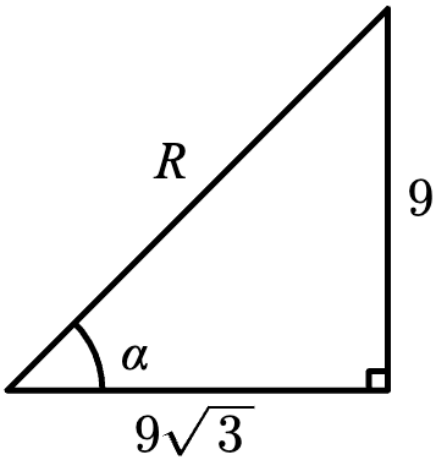
$$R = \sqrt{5^2 + (5\sqrt{3})^2} = 10$$

$$\alpha = \arctan\left(\frac{5\sqrt{3}}{5}\right) = 60^\circ$$

$$\therefore 5\sin x + 5\sqrt{3}\cos x = 10\sin(x+60^\circ)$$

6 c)

$$R\sin(x+\alpha) \equiv R(\sin x\cos\alpha + \cos x\sin\alpha)$$
$$\equiv R\cos\alpha\sin x + R\sin\alpha\cos x$$
$$9\sqrt{3}\sin x + 9\cos x \equiv R\cos\alpha\sin x + R\sin\alpha\cos x$$
$$\therefore 9\sqrt{3} = R\cos\alpha \text{ and } 9 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{9\sqrt{3}}{R} \text{ and } \sin\alpha = \frac{9}{R}$$



$$R = \sqrt{(9\sqrt{3})^2 + 9^2} = 18$$
$$\alpha = \arctan\left(\frac{9}{9\sqrt{3}}\right) = 30^\circ$$
$$\therefore 9\sqrt{3}\sin x + 9\cos x = 18\sin(x+30^\circ)$$

6 d)

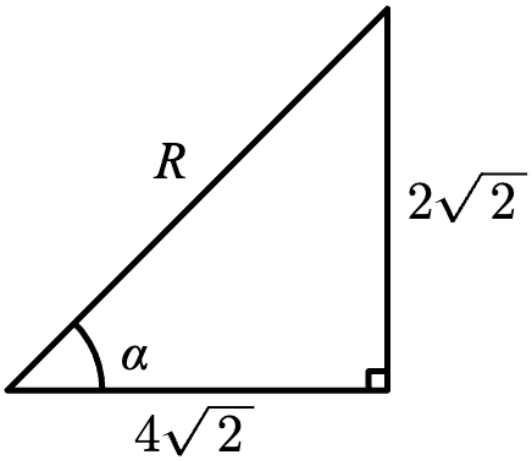
$$R\sin(x+\alpha) \equiv R(\sin x\cos\alpha + \cos x\sin\alpha)$$

$$\equiv R\cos\alpha\sin x + R\sin\alpha\cos x$$

$$4\sqrt{2}\sin x + 2\sqrt{2}\cos x \equiv R\cos\alpha\sin x + R\sin\alpha\cos x$$

$$\therefore 4\sqrt{2} = R\cos\alpha \text{ and } 2\sqrt{2} = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{4\sqrt{2}}{R} \text{ and } \sin\alpha = \frac{2\sqrt{2}}{R}$$



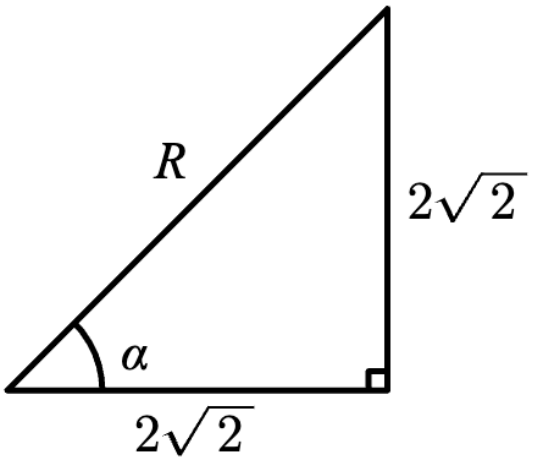
$$R = \sqrt{(4\sqrt{2})^2 + (2\sqrt{2})^2} = 2\sqrt{10}$$

$$\alpha = \arctan\left(\frac{2\sqrt{2}}{4\sqrt{2}}\right) = 26.6^\circ \text{ to 3sf}$$

$$\therefore 4\sqrt{2}\sin x + 2\sqrt{2}\cos x = 2\sqrt{10}\sin(x + 26.6^\circ)$$

7 a)

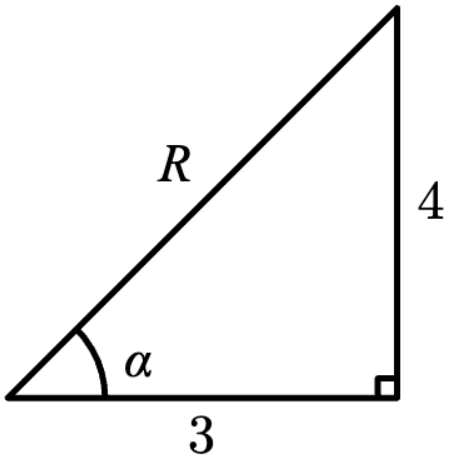
$$R\sin(x-\alpha) \equiv R(\sin x\cos\alpha - \cos x\sin\alpha)$$
$$\equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$2\sqrt{2}\sin x - 2\sqrt{2}\cos x \equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$\therefore 2\sqrt{2} = R\cos\alpha \text{ and } 2\sqrt{2} = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{2\sqrt{2}}{R} \text{ and } \sin\alpha = \frac{2\sqrt{2}}{R}$$



$$R = \sqrt{(2\sqrt{2})^2 + (2\sqrt{2})^2} = 4$$
$$\alpha = \arctan\left(\frac{2\sqrt{2}}{2\sqrt{2}}\right) = \frac{\pi}{4}$$
$$\therefore 2\sqrt{2}\sin x - 2\sqrt{2}\cos x = 4\sin\left(x - \frac{\pi}{4}\right)$$

7 b)

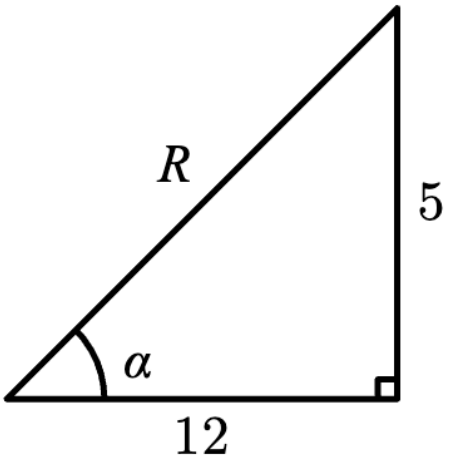
$$R\sin(x-\alpha) \equiv R(\sin x\cos\alpha - \cos x\sin\alpha)$$
$$\equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$3\sin x - 4\cos x \equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$\therefore 3 = R\cos\alpha \text{ and } 4 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{3}{R} \text{ and } \sin\alpha = \frac{4}{R}$$



$$R = \sqrt{3^2 + 4^2} = 5$$
$$\alpha = \arctan\left(\frac{4}{3}\right) = 0.927 \text{ to 3sf}$$
$$\therefore 3\sin x - 4\cos x = 5\sin(x - 0.927)$$

7 c)

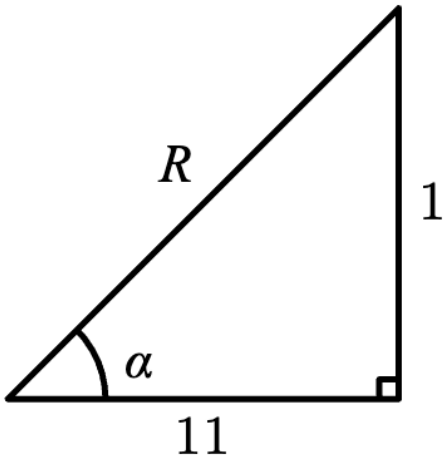
$$R\sin(x-\alpha) \equiv R(\sin x\cos\alpha - \cos x\sin\alpha)$$
$$\equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$12\sin x - 5\cos x \equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$\therefore 12 = R\cos\alpha \text{ and } 5 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{12}{R} \text{ and } \sin\alpha = \frac{5}{R}$$



$$R = \sqrt{12^2 + 5^2} = 13$$
$$\alpha = \arctan\left(\frac{5}{12}\right) = 0.395 \text{ to 3sf}$$
$$\therefore 12\sin x - 5\cos x = 13\sin(x - 0.395)$$

7 d)

$$R\sin(x-\alpha) \equiv R(\sin x\cos\alpha - \cos x\sin\alpha)$$
$$\equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$11\sin x - \cos x \equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$\therefore 11 = R\cos\alpha \text{ and } 1 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{11}{R} \text{ and } \sin\alpha = \frac{1}{R}$$



$$R = \sqrt{11^2 + 1^2} = \sqrt{122}$$
$$\alpha = \arctan\left(\frac{1}{11}\right) = 0.0907 \text{ to 3sf}$$
$$\therefore 11\sin x - \cos x = \sqrt{122}\sin(x - 0.0907)$$

8 a)

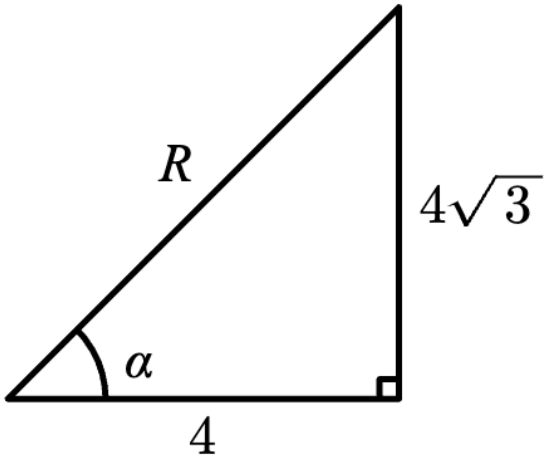
$$R\cos(x+\alpha) \equiv R(\cos x \cos\alpha - \sin x \sin\alpha)$$

$$\equiv R\cos\alpha\cos x - R\sin\alpha\sin x$$

$$4\cos x - 4\sqrt{3}\sin x \equiv R\cos\alpha\cos x - R\sin\alpha\sin x$$

$$\therefore 4 = R\cos\alpha \text{ and } 4\sqrt{3} = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{4}{R} \text{ and } \sin\alpha = \frac{4\sqrt{3}}{R}$$



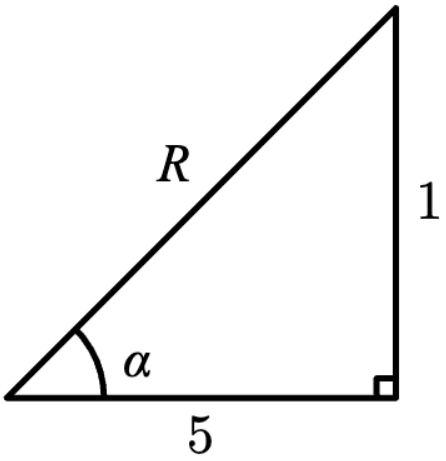
$$R = \sqrt{4^2 + (4\sqrt{3})^2} = 8$$

$$\alpha = \arctan\left(\frac{4\sqrt{3}}{4}\right) = 60^\circ$$

$$\therefore 4\cos x - 4\sqrt{3}\sin x = 8\cos(x + 60^\circ)$$

8 b)

$$R\cos(x+\alpha) \equiv R(\cos x \cos\alpha - \sin x \sin\alpha)$$
$$\equiv R\cos\alpha\cos x - R\sin\alpha\sin x$$
$$5\cos x - \sin x \equiv R\cos\alpha\cos x - R\sin\alpha\sin x$$
$$\therefore 5 = R\cos\alpha \text{ and } 1 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{5}{R} \text{ and } \sin\alpha = \frac{1}{R}$$



$$R = \sqrt{5^2 + 1^2} = \sqrt{26}$$
$$\alpha = \arctan\left(\frac{1}{5}\right) = 11.3^\circ \text{ to 3sf}$$
$$\therefore 5\cos x - \sin x = \sqrt{26}\cos(x + 11.3^\circ)$$

8 c)

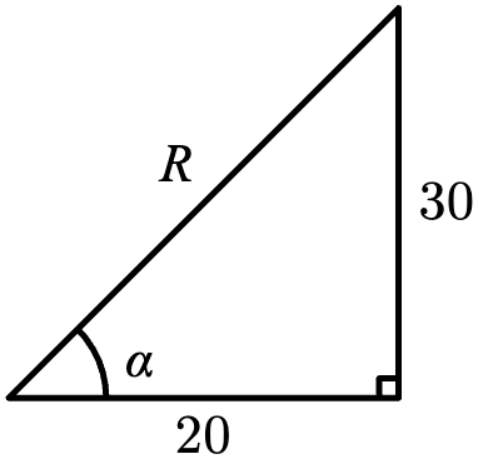
$$R\cos(x+\alpha) \equiv R(\cos x \cos\alpha - \sin x \sin\alpha)$$

$$\equiv R\cos\alpha\cos x - R\sin\alpha\sin x$$

$$20\cos x - 30\sin x \equiv R\cos\alpha\cos x - R\sin\alpha\sin x$$

$$\therefore 20 = R\cos\alpha \text{ and } 30 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{20}{R} \text{ and } \sin\alpha = \frac{30}{R}$$



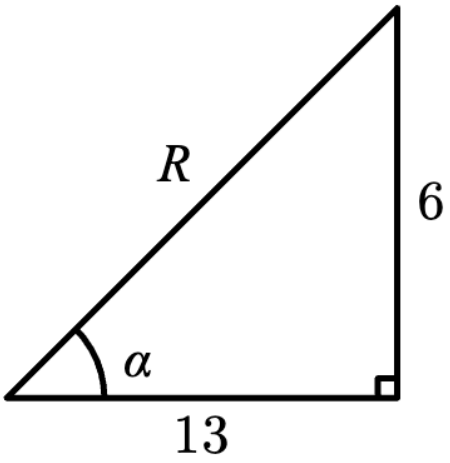
$$R = \sqrt{20^2 + 30^2} = 10\sqrt{13}$$

$$\alpha = \arctan\left(\frac{30}{20}\right) = 56.3^\circ \text{ to 3sf}$$

$$\therefore 20\cos x - 30\sin x = 10\sqrt{13}\cos(x + 56.3^\circ)$$

8 d)

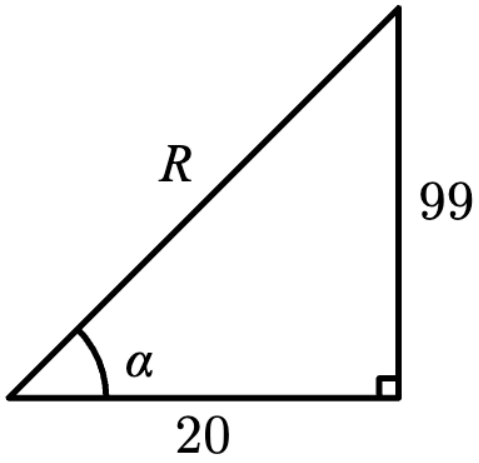
$$R\cos(x+\alpha) \equiv R(\cos x \cos\alpha - \sin x \sin\alpha)$$
$$\equiv R\cos\alpha\cos x - R\sin\alpha\sin x$$
$$13\cos x - 6\sin x \equiv R\cos\alpha\cos x - R\sin\alpha\sin x$$
$$\therefore 13 = R\cos\alpha \text{ and } 6 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{13}{R} \text{ and } \sin\alpha = \frac{6}{R}$$



$$R = \sqrt{13^2 + 6^2} = \sqrt{205}$$
$$\alpha = \arctan\left(\frac{6}{13}\right) = 24.8^\circ \text{ to 3sf}$$
$$\therefore 13\cos x - 6\sin x = \sqrt{205}\cos(x + 24.8^\circ)$$

9 a)

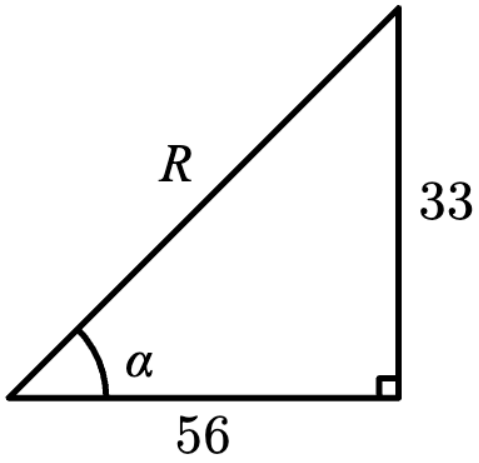
$$R\cos(x-\alpha) \equiv R(\cos x\cos\alpha + \sin x\sin\alpha)$$
$$\equiv R\cos\alpha\cos x + R\sin\alpha\sin x$$
$$20\cos x + 99\sin x \equiv R\cos\alpha\cos x + R\sin\alpha\sin x$$
$$\therefore 20 = R\cos\alpha \text{ and } 99 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{20}{R} \text{ and } \sin\alpha = \frac{99}{R}$$



$$R = \sqrt{20^2 + 99^2} = 101$$
$$\alpha = \arctan\left(\frac{99}{20}\right) = 1.37 \text{ to 3sf}$$
$$\therefore 20\cos x + 99\sin x = 101\cos(x - 1.37)$$

9 b)

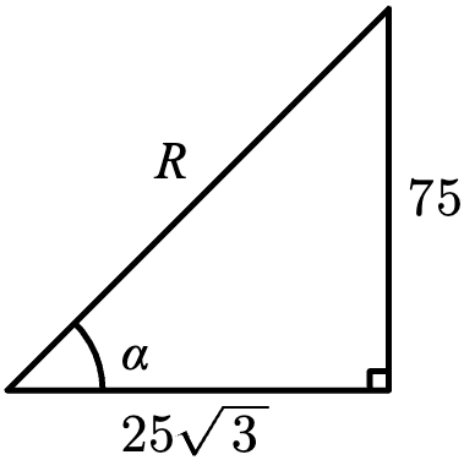
$$R\cos(x-\alpha) \equiv R(\cos x\cos\alpha + \sin x\sin\alpha)$$
$$\equiv R\cos\alpha\cos x + R\sin\alpha\sin x$$
$$56\cos x + 33\sin x \equiv R\cos\alpha\cos x + R\sin\alpha\sin x$$
$$\therefore 56 = R\cos\alpha \text{ and } 33 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{56}{R} \text{ and } \sin\alpha = \frac{33}{R}$$



$$R = \sqrt{56^2 + 33^2} = 65$$
$$\alpha = \arctan\left(\frac{33}{56}\right) = 0.533 \text{ to 3sf}$$
$$\therefore 56\cos x + 33\sin x = 65\cos(x - 0.533)$$

9 c)

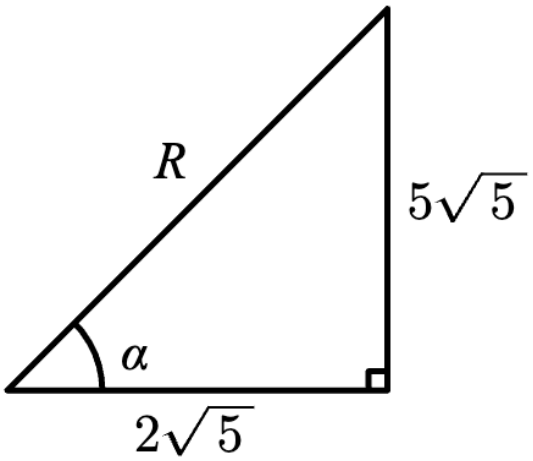
$$R\cos(x-\alpha) \equiv R(\cos x\cos\alpha + \sin x\sin\alpha)$$
$$\equiv R\cos\alpha\cos x + R\sin\alpha\sin x$$
$$25\sqrt{3}\cos x + 75\sin x \equiv R\cos\alpha\cos x + R\sin\alpha\sin x$$
$$\therefore 25\sqrt{3} = R\cos\alpha \text{ and } 75 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{25\sqrt{3}}{R} \text{ and } \sin\alpha = \frac{75}{R}$$



$$R = \sqrt{(25\sqrt{3})^2 + 75^2} = 50\sqrt{3}$$
$$\alpha = \arctan\left(\frac{75}{25\sqrt{3}}\right) = \frac{\pi}{3}$$
$$\therefore 25\sqrt{3}\cos x + 75\sin x = 50\sqrt{3}\cos\left(x - \frac{\pi}{3}\right)$$

9 d)

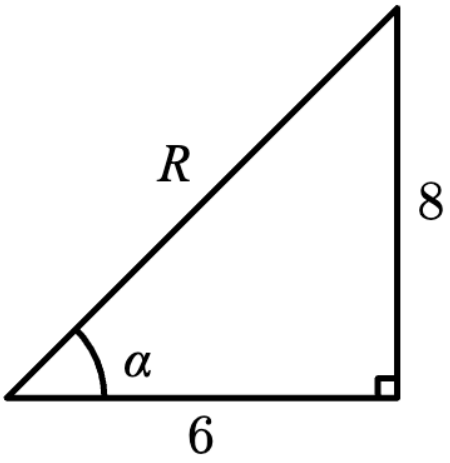
$$R\cos(x-\alpha) \equiv R(\cos x\cos\alpha + \sin x\sin\alpha)$$
$$\equiv R\cos\alpha\cos x + R\sin\alpha\sin x$$
$$2\sqrt{5}\cos x + 5\sqrt{5}\sin x \equiv R\cos\alpha\cos x + R\sin\alpha\sin x$$
$$\therefore 2\sqrt{5} = R\cos\alpha \text{ and } 5\sqrt{5} = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{2\sqrt{5}}{R} \text{ and } \sin\alpha = \frac{5\sqrt{5}}{R}$$



$$R = \sqrt{(2\sqrt{5})^2 + (5\sqrt{5})^2} = \sqrt{145}$$
$$\alpha = \arctan\left(\frac{5\sqrt{5}}{2\sqrt{5}}\right) = 1.19 \text{ to 3sf}$$
$$\therefore 2\sqrt{5}\cos x + 5\sqrt{5}\sin x = \sqrt{145}\cos(x - 1.19)$$

10 a)

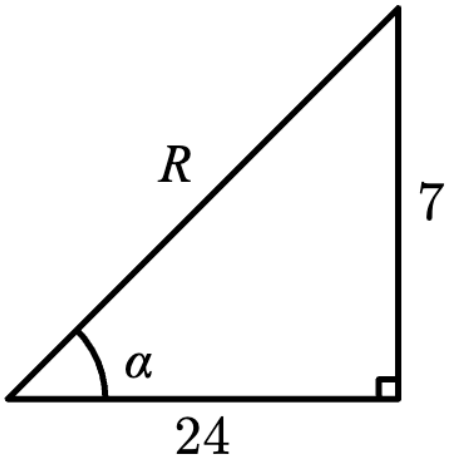
$$R\cos(\theta+\alpha) \equiv R(\cos\theta\cos\alpha - \sin\theta\sin\alpha)$$
$$\equiv R\cos\alpha\cos\theta - R\sin\alpha\sin\theta$$
$$6\cos\theta - 8\sin\theta \equiv R\cos\alpha\cos\theta - R\sin\alpha\sin\theta$$
$$\therefore 6 = R\cos\alpha \text{ and } 8 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{6}{R} \text{ and } \sin\alpha = \frac{8}{R}$$



$$R = \sqrt{6^2+8^2} = 10$$
$$\alpha = \arctan\left(\frac{8}{6}\right) = 53.1^\circ \text{ to 3sf}$$
$$\therefore 6\cos\theta - 8\sin\theta = 10\cos(\theta + 53.1^\circ)$$

10 b)

$$R\sin(\theta+\alpha) \equiv R(\sin\theta\cos\alpha + \cos\theta\sin\alpha)$$
$$\equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$
$$24\sin\theta + 7\cos\theta \equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$
$$\therefore 24 = R\cos\alpha \text{ and } 7 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{24}{R} \text{ and } \sin\alpha = \frac{7}{R}$$



$$R = \sqrt{24^2 + 7^2} = 25$$
$$\alpha = \arctan\left(\frac{7}{24}\right) = 16.3^\circ \text{ to 3sf}$$
$$\therefore 24\sin\theta + 7\cos\theta = 25\sin(\theta + 16.3^\circ)$$

Alternatively:

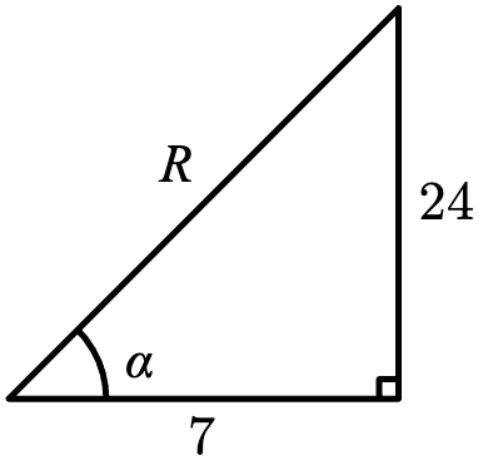
$$R\cos(\theta-\alpha) \equiv R(\cos\theta\cos\alpha+\sin\theta\sin\alpha)$$

$$\equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$7\cos\theta + 24\sin\theta \equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$\therefore 7 = R\cos\alpha \text{ and } 24 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{7}{R} \text{ and } \sin\alpha = \frac{24}{R}$$



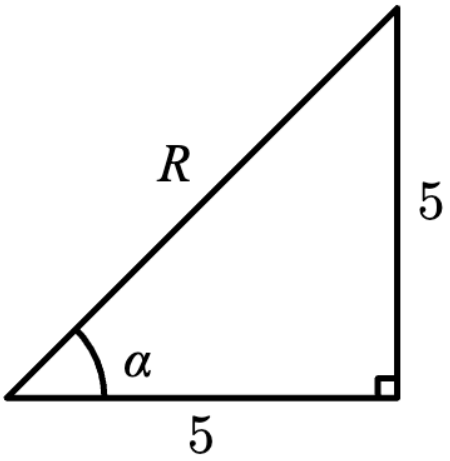
$$R = \sqrt{7^2+24^2} = 25$$

$$\alpha = \arctan\left(\frac{24}{7}\right) = 73.7^\circ \text{ to 3sf}$$

$$\therefore 7\cos\theta + 24\sin\theta = 25\cos(\theta - 73.7^\circ)$$

10 c)

$$R\sin(\theta-\alpha) \equiv R(\sin\theta\cos\alpha - \cos\theta\sin\alpha)$$
$$\equiv R\cos\alpha\sin\theta - R\sin\alpha\cos\theta$$
$$5\sin\theta - 5\cos\theta \equiv R\cos\alpha\sin\theta - R\sin\alpha\cos\theta$$
$$\therefore 5 = R\cos\alpha \text{ and } 5 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{5}{R} \text{ and } \sin\alpha = \frac{5}{R}$$



$$R = \sqrt{5^2+5^2} = 5\sqrt{2}$$
$$\alpha = \arctan\left(\frac{5}{5}\right) = 45^\circ$$
$$\therefore 5\sin\theta - 5\cos\theta = 5\sqrt{2}\sin(\theta - 45^\circ)$$

10 d)

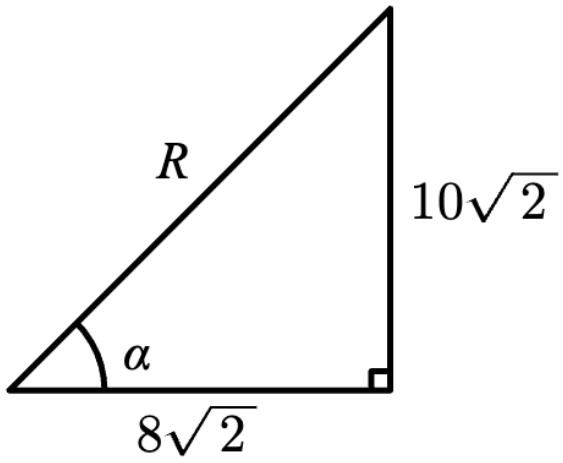
$$R\sin(\theta+\alpha) \equiv R(\sin\theta\cos\alpha + \cos\theta\sin\alpha)$$

$$\equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$

$$8\sqrt{2}\sin\theta + 10\sqrt{2}\cos\theta \equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$

$$\therefore 8\sqrt{2} = R\cos\alpha \text{ and } 10\sqrt{2} = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{8\sqrt{2}}{R} \text{ and } \sin\alpha = \frac{10\sqrt{2}}{R}$$



$$R = \sqrt{(8\sqrt{2})^2 + (10\sqrt{2})^2} = 2\sqrt{82}$$

$$\alpha = \arctan\left(\frac{10\sqrt{2}}{8\sqrt{2}}\right) = 51.3^\circ \text{ to 3sf}$$

$$\therefore 8\sqrt{2}\sin\theta + 10\sqrt{2}\cos\theta = 2\sqrt{82}\sin(\theta + 51.3^\circ)$$

Alternatively:

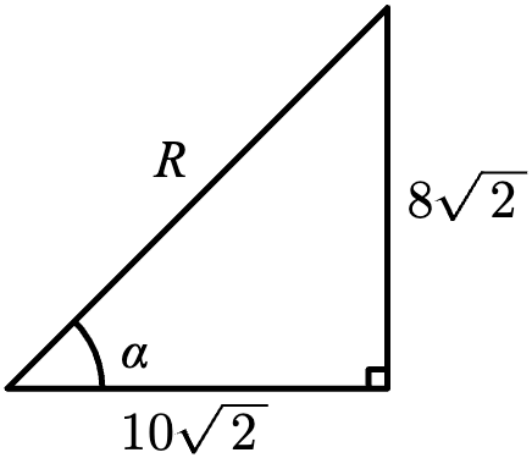
$$R\cos(\theta-\alpha) \equiv R(\cos\theta\cos\alpha+\sin\theta\sin\alpha)$$

$$\equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$10\sqrt{2}\cos\theta + 8\sqrt{2}\sin\theta \equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$\therefore 10\sqrt{2} = R\cos\alpha \text{ and } 8\sqrt{2} = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{10\sqrt{2}}{R} \text{ and } \sin\alpha = \frac{8\sqrt{2}}{R}$$



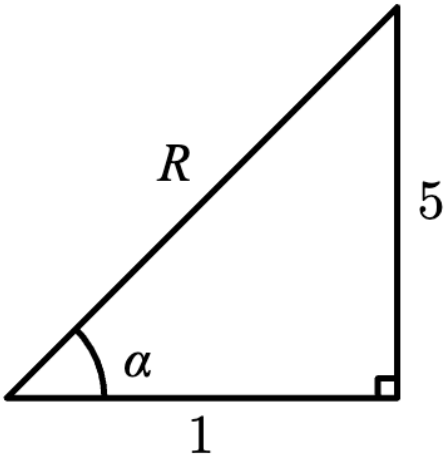
$$R = \sqrt{(10\sqrt{2})^2 + (8\sqrt{2})^2} = 2\sqrt{82}$$

$$\alpha = \arctan\left(\frac{8\sqrt{2}}{10\sqrt{2}}\right) = 38.7^\circ \text{ to 3sf}$$

$$\therefore 10\sqrt{2}\cos\theta + 8\sqrt{2}\sin\theta = 2\sqrt{82}\cos(x - 38.7^\circ)$$

11 a)

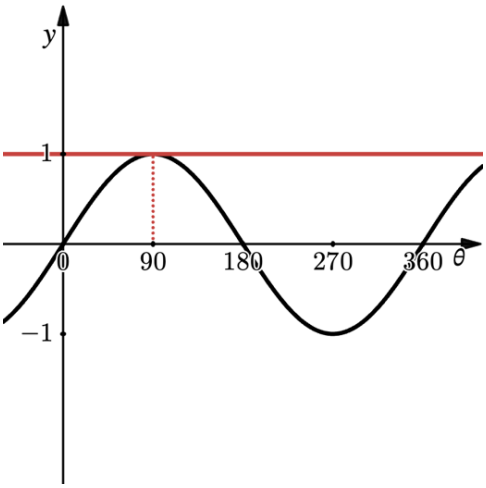
$$R\sin(\theta+\alpha) \equiv R(\sin\theta\cos\alpha + \cos\theta\sin\alpha)$$
$$\equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$
$$\sin\theta + 5\cos\theta \equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$
$$\therefore 1 = R\cos\alpha \text{ and } 5 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{1}{R} \text{ and } \sin\alpha = \frac{5}{R}$$



$$R = \sqrt{1^2+5^2} = \sqrt{26}$$
$$\alpha = \arctan\left(\frac{5}{1}\right) = 78.690\ldots^\circ$$
$$\therefore \sin\theta + 5\cos\theta = \sqrt{26}\sin(\theta + 78.690\ldots^\circ)$$

Maximum is $\sqrt{26}$.
Solve $\sqrt{26}\sin(\theta + 78.690\ldots^\circ) = \sqrt{26}$

$$\Rightarrow \sin(\theta + 78.690\ldots^\circ) = 1$$



$$\theta + 78.690\ldots^\circ = 90^\circ$$

Apply transformation:

$$\theta = 11.309\ldots^\circ = 11.3^\circ \text{ to 3sf}$$

$\therefore$ Maximum $= \sqrt{26}$ when $\theta = 11.3^\circ$.

Alternatively:

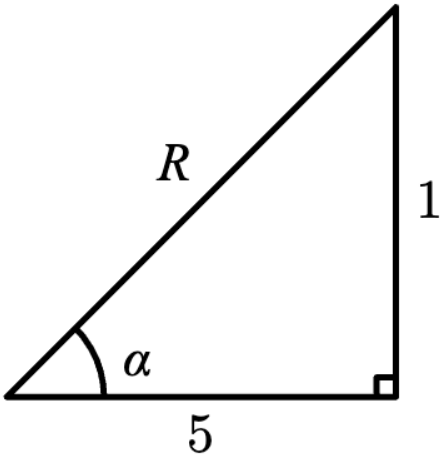
$$R\cos(\theta - \alpha) \equiv R(\cos\theta\cos\alpha + \sin\theta\sin\alpha)$$

$$\equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$5\cos\theta + \sin\theta \equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$\therefore 5 = R\cos\alpha \text{ and } 1 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{5}{R} \text{ and } \sin\alpha = \frac{1}{R}$$



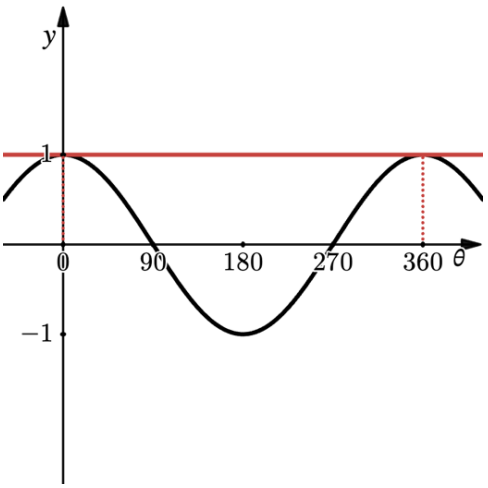
$$R = \sqrt{5^2 + 1^2} = \sqrt{26}$$

$$\alpha = \arctan\left(\frac{1}{5}\right) = 11.309\ldots^\circ$$

$$\therefore 5\cos\theta + \sin\theta = \sqrt{26}\cos(x - 11.309\ldots^\circ)$$

Maximum is $\sqrt{26}$.
Solve $\sqrt{26}\cos(\theta - 11.309\ldots^\circ) = \sqrt{26}$

$$\Rightarrow \cos(\theta - 11.309\ldots^\circ) = 1$$



$$\theta - 11.309\ldots^\circ = 0^\circ$$

Apply transformation:

$$\theta = 11.309\ldots^\circ = 11.3^\circ \text{ to 3sf}$$

$\therefore$ Maximum $= \sqrt{26}$ when $\theta = 11.3^\circ$.

11 b)

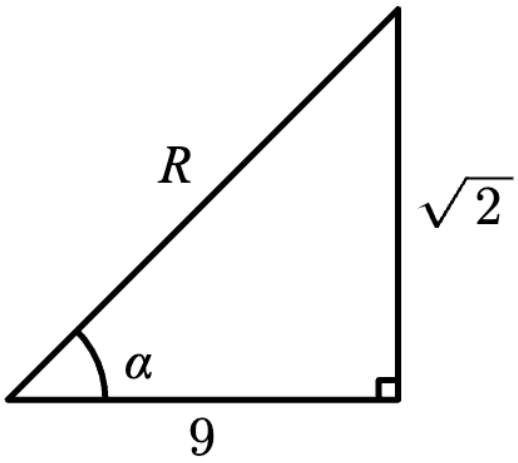
$$R\cos(\theta+\alpha) \equiv R(\cos\theta\cos\alpha - \sin\theta\sin\alpha)$$

$$\equiv R\cos\alpha\cos\theta - R\sin\alpha\sin\theta$$

$$9\cos\theta - \sqrt{2}\sin\theta \equiv R\cos\alpha\cos\theta - R\sin\alpha\sin\theta$$

$$\therefore 9 = R\cos\alpha \text{ and } \sqrt{2} = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{9}{R} \text{ and } \sin\alpha = \frac{\sqrt{2}}{R}$$



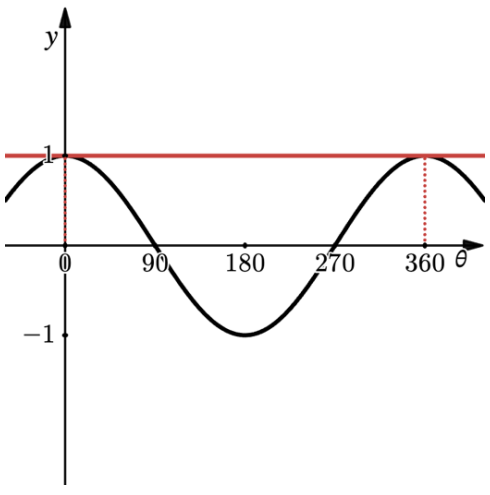
$$R = \sqrt{9^2 + (\sqrt{2})^2} = \sqrt{83}$$

$$\alpha = \arctan\left(\frac{\sqrt{2}}{9}\right) = 8.930\ldots^\circ$$

$$\therefore 9\cos\theta - \sqrt{2}\sin\theta = \sqrt{83}\cos(\theta + 8.930\ldots^\circ)$$

Maximum is $\sqrt{83}$.
Solve $\sqrt{83}\cos(\theta + 8.930\ldots^\circ) = \sqrt{26}$

$$\Rightarrow \cos(\theta + 8.930\ldots^\circ) = 1$$



$$\theta + 8.930\ldots^\circ = 0^\circ$$

Apply transformation:

$$\theta = -8.930\ldots^\circ$$

Add period:

$$\theta = -8.930\ldots^\circ + 360^\circ = 351.069\ldots^\circ = 351^\circ \text{ to 3sf}$$

$\therefore$ Maximum $= \sqrt{83}$ when $\theta = 351^\circ$.

11 c)

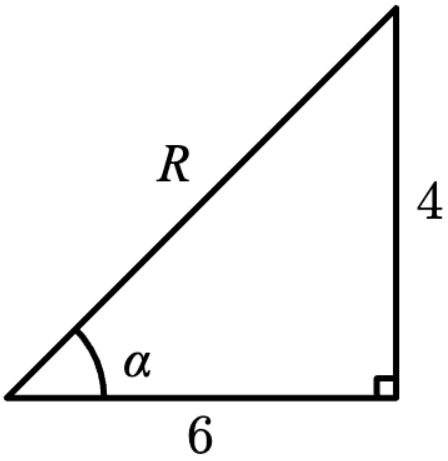
$$R\sin(\theta - \alpha) \equiv R(\sin\theta\cos\alpha - \cos\theta\sin\alpha)$$

$$\equiv R\cos\alpha\sin\theta - R\sin\alpha\cos\theta$$

$$6\sin\theta - 4\cos\theta \equiv R\cos\alpha\sin\theta - R\sin\alpha\cos\theta$$

$$\therefore 6 = R\cos\alpha \text{ and } 4 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{6}{R} \text{ and } \sin\alpha = \frac{4}{R}$$



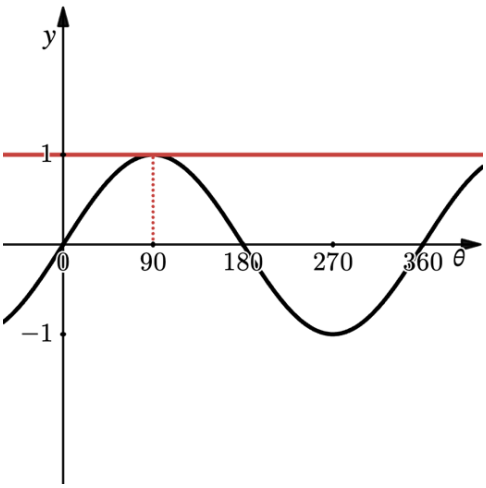
$$R = \sqrt{6^2 + 4^2} = 2\sqrt{13}$$

$$\alpha = \arctan\left(\frac{4}{6}\right) = 33.690\ldots^\circ$$

$$\therefore 6\sin\theta - 4\cos\theta = 2\sqrt{13}\sin(\theta - 33.690\ldots^\circ)$$

Maximum is $2\sqrt{13}$.
Solve $2\sqrt{13}\sin(\theta - 33.690\ldots^\circ) = 2\sqrt{13}$

$$\Rightarrow \sin(\theta - 33.690\ldots^\circ) = 1$$



$$\theta - 33.690\ldots^\circ = 90^\circ$$

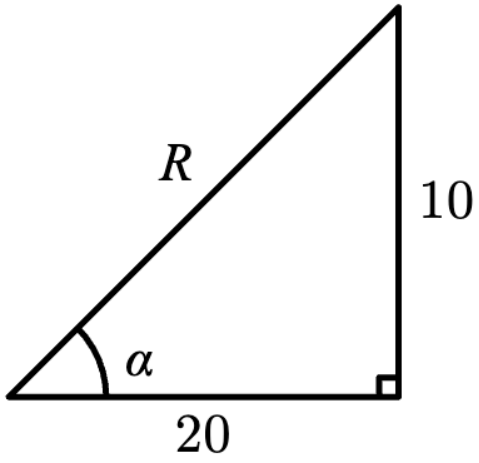
Apply transformation:

$$\theta = 123.690\ldots^\circ = 124^\circ \text{ to 3sf}$$

$\therefore$ Maximum $= 2\sqrt{13}$ when $\theta = 124^\circ$.

11 d)

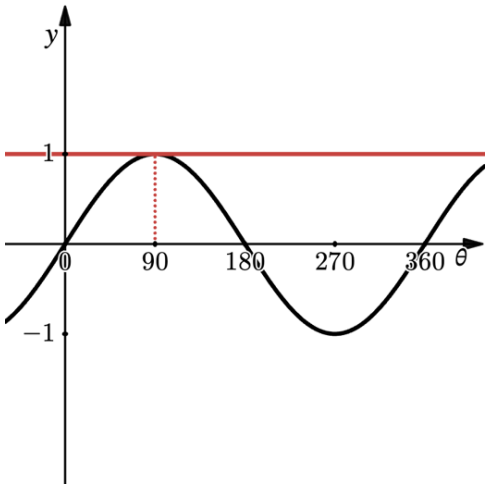
$$R\sin(\theta+\alpha) \equiv R(\sin\theta\cos\alpha + \cos\theta\sin\alpha)$$
$$\equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$
$$20\sin\theta + 10\cos\theta \equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$
$$\therefore 20 = R\cos\alpha \text{ and } 10 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{20}{R} \text{ and } \sin\alpha = \frac{10}{R}$$



$$R = \sqrt{20^2 + 10^2} = 10\sqrt{5}$$
$$\alpha = \arctan\left(\frac{10}{20}\right) = 26.565\ldots^\circ$$
$$\therefore 20\sin\theta + 10\cos\theta = 10\sqrt{5}\sin(\theta + 26.565\ldots^\circ)$$

Maximum is $10\sqrt{5}$.
Solve $10\sqrt{5}\sin(\theta + 26.565\ldots^\circ) = 10\sqrt{5}$

$$\Rightarrow \sin(\theta + 26.565\ldots^\circ) = 1$$



$$\theta + 26.565\ldots^\circ = 90^\circ$$

Apply transformation:

$$\theta = 63.434\ldots^\circ = 63.4^\circ \text{ to 3sf}$$

$\therefore$ Maximum $= 10\sqrt{5}$ when $\theta = 63.4^\circ$.

Alternatively:

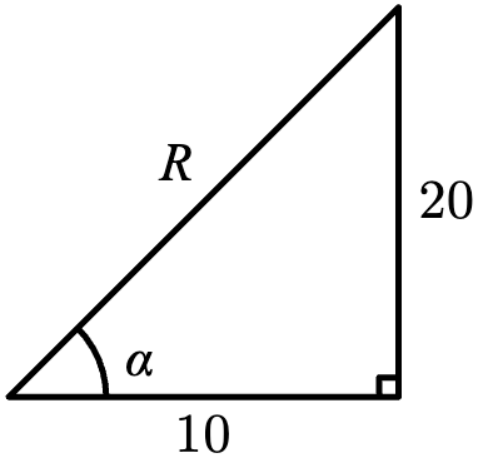
$$R\cos(\theta - \alpha) \equiv R(\cos\theta\cos\alpha + \sin\theta\sin\alpha)$$

$$\equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$10\cos\theta + 20\sin\theta \equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$\therefore 10 = R\cos\alpha \text{ and } 20 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{10}{R} \text{ and } \sin\alpha = \frac{20}{R}$$



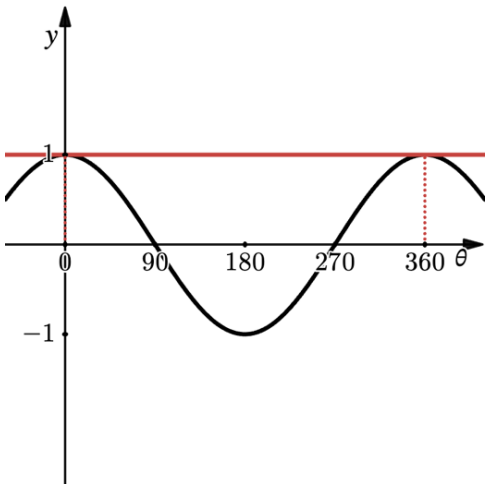
$$R = \sqrt{10^2 + 20^2} = 10\sqrt{5}$$

$$\alpha = \arctan\left(\frac{20}{10}\right) = 63.434\ldots^\circ$$

$$\therefore 10\cos\theta + 20\sin\theta = 10\sqrt{5}\cos(\theta - 63.434\ldots^\circ)$$

Maximum is $10\sqrt{5}$.
Solve $10\sqrt{5}\cos(\theta - 63.434\ldots^\circ) = 10\sqrt{5}$

$$\Rightarrow \cos(\theta - 63.434\ldots^\circ) = 1$$



$$\theta - 63.434\ldots^\circ = 0^\circ$$

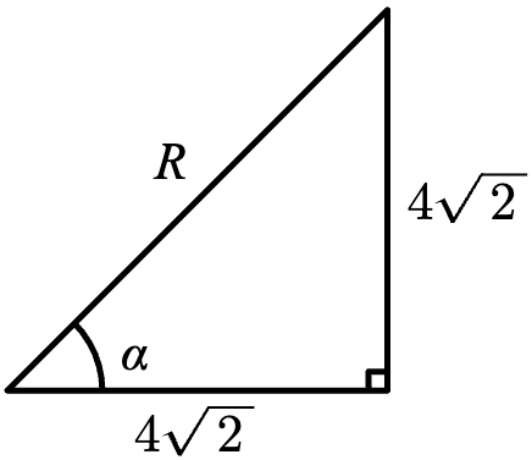
Apply transformation:

$$\theta = 63.434\ldots^\circ = 63.4^\circ \text{ to 3sf}$$

$\therefore$ Maximum $= 10\sqrt{5}$ when $\theta = 63.4^\circ$.

12 a)

$$R\sin(\theta-\alpha) \equiv R(\sin\theta\cos\alpha - \cos\theta\sin\alpha)$$
$$\equiv R\cos\alpha\sin\theta - R\sin\alpha\cos\theta$$
$$4\sqrt{2}\sin\theta - 4\sqrt{2}\cos\theta \equiv R\cos\alpha\sin\theta - R\sin\alpha\cos\theta$$
$$\therefore 4\sqrt{2} = R\cos\alpha \text{ and } 4\sqrt{2} = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{4\sqrt{2}}{R} \text{ and } \sin\alpha = \frac{4\sqrt{2}}{R}$$



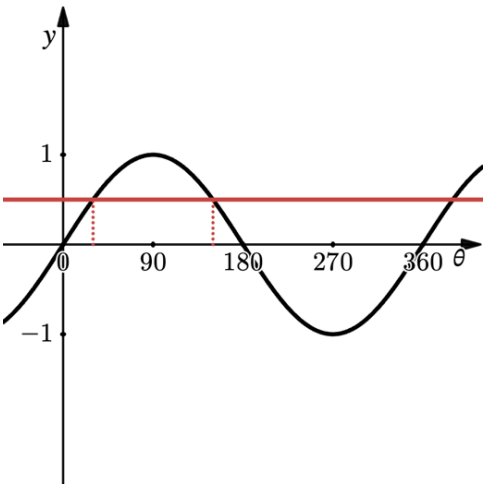
$$R = \sqrt{(4\sqrt{2})^2 + (4\sqrt{2})^2} = 8$$
$$\alpha = \arctan\left(\frac{4\sqrt{2}}{4\sqrt{2}}\right) = 45^\circ$$
$$\therefore 4\sqrt{2}\sin\theta - 4\sqrt{2}\cos\theta = 8\sin(\theta - 45^\circ)$$

$4\sqrt{2}\sin\theta - 4\sqrt{2}\cos\theta = 7$ for $0^\circ < \theta < 360^\circ$

$$\Rightarrow 8\sin(\theta - 45^\circ) = 7 \Rightarrow \sin(\theta - 45^\circ) = \frac{7}{8}$$

Find first value between 0° and 360°:

$$\theta - 45^\circ = \arcsin\frac{7}{8} = 61.044\ldots^\circ$$



Find second value between $0°$ and $360°$:

$$\theta - 45° = 180° - 61.044\ldots° = 118.955\ldots°$$

Apply transformation:

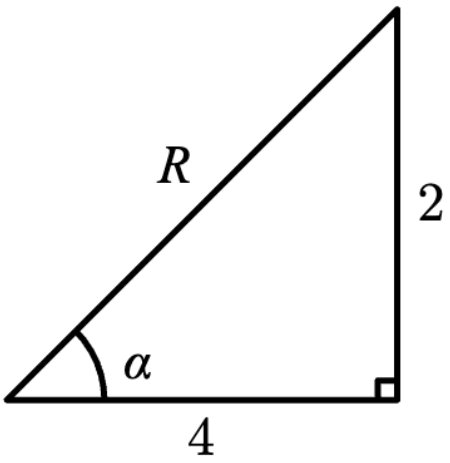
$$\theta - 45° = 61.044\ldots° \Rightarrow \theta = 106.044\ldots° \quad \text{(in range)}$$

$$\theta - 45° = 118.955\ldots° \Rightarrow \theta = 163.955\ldots° \quad \text{(in range)}$$

$\theta = \{106°, 164°\}$ to 3sf.

12 b)

$$R\sin(\theta + \alpha) \equiv R(\sin\theta\cos\alpha + \cos\theta\sin\alpha)$$
$$\equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$
$$4\sin\theta + 2\cos\theta \equiv R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$
$$\therefore 4 = R\cos\alpha \text{ and } 2 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{4}{R} \text{ and } \sin\alpha = \frac{2}{R}$$



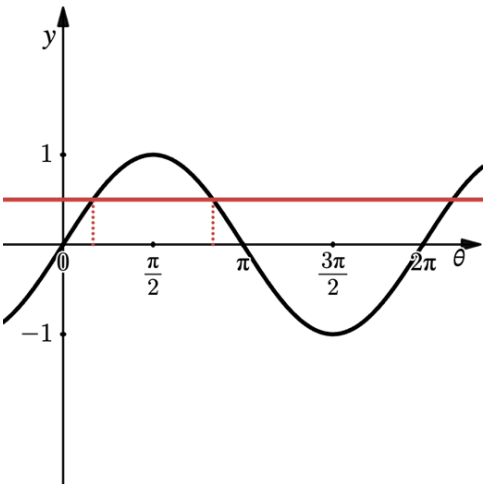
$$R = \sqrt{4^2 + 2^2} = 2\sqrt{5}$$
$$\alpha = \arctan\left(\frac{2}{4}\right) = 0.4636\ldots$$
$$\therefore 4\sin\theta + 2\cos\theta = 2\sqrt{5}\sin(\theta + 0.4636\ldots)$$

$4\sin\theta + 2\cos\theta = 3$ for $0 < \theta < 2\pi$

$$\Rightarrow 2\sqrt{5}\sin(\theta + 0.4636\ldots) = 3$$
$$\Rightarrow \sin(\theta + 0.4636\ldots) = \frac{3}{2\sqrt{5}}$$

Find first value between 0 and 2π:

$$\theta + 0.4636\ldots = \arcsin\frac{3}{2\sqrt{5}} = 0.7353\ldots$$



Find second value between 0 and 2π:

$$\theta + 0.4636\ldots = \pi - 0.7353\ldots = 2.4062\ldots$$

Apply transformation:

$$\theta + 0.4636\ldots = 0.7353\ldots \Rightarrow \theta = 0.2716\ldots \quad \text{(in range)}$$

$$\theta + 0.4636\ldots = 2.4062\ldots \Rightarrow \theta = 1.9426\ldots \quad \text{(in range)}$$

$\theta = \{0.272, 1.94\}$ to 3sf.

Alternatively:

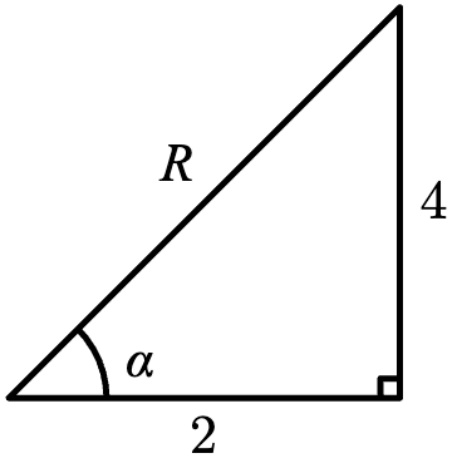
$$R\cos(\theta-\alpha) \equiv R(\cos\theta\cos\alpha + \sin\theta\sin\alpha)$$

$$\equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$2\cos\theta + 4\sin\theta \equiv R\cos\alpha\cos\theta + R\sin\alpha\sin\theta$$

$$\therefore 2 = R\cos\alpha \text{ and } 4 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{2}{R} \text{ and } \sin\alpha = \frac{4}{R}$$



$$R = \sqrt{2^2+4^2} = 2\sqrt{5}$$

$$\alpha = \arctan\left(\frac{4}{2}\right) = 1.1071\ldots$$

$$\therefore 2\cos\theta + 4\sin\theta = 2\sqrt{5}\cos(\theta - 1.1071\ldots)$$

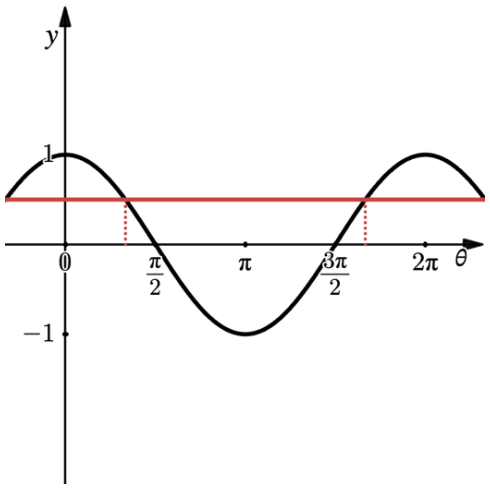
$2\cos\theta + 4\sin\theta = 3$ for $0 < \theta < 2\pi$

$$\Rightarrow 2\sqrt{5}\cos(\theta - 1.1071\ldots) = 3$$

$$\Rightarrow \cos(\theta - 1.1071\ldots) = \frac{3}{2\sqrt{5}}$$

Find first value between 0 and 2π:

$$\theta - 1.1071\ldots = \arccos\frac{3}{2\sqrt{5}} = 0.8354\ldots$$



Find second value between 0 and 2π:

$$\theta - 1.1071\ldots = 2\pi - 0.8354\ldots = 5.4477\ldots$$

Apply transformation:

$$\theta - 1.1071\ldots = 0.8354\ldots \Rightarrow \theta = 1.9426\ldots \quad \text{(in range)}$$

$$\theta - 1.1071\ldots = 5.4477\ldots \Rightarrow \theta = 6.5548\ldots \quad \text{(not in range)}$$

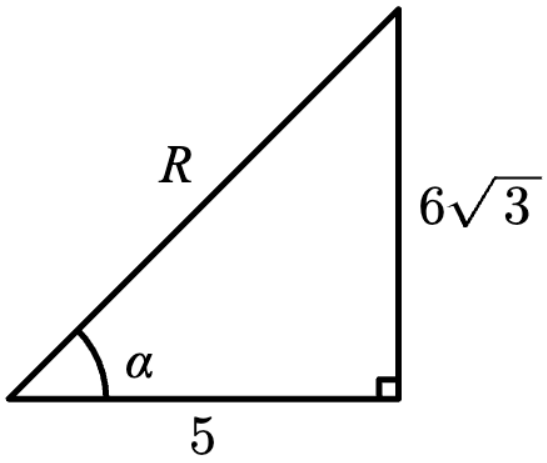
Subtract period:

$$\theta = 6.5548\ldots - 2\pi = 0.2716\ldots \quad \text{(in range)}$$

$\theta = \{0.272, 1.94\}$ to 3sf.

12 c)

$$R\sin(3\theta + \alpha) \equiv R(\sin 3\theta \cos\alpha + \cos 3\theta \sin\alpha)$$
$$\equiv R\cos\alpha \sin 3\theta + R\sin\alpha \cos 3\theta$$
$$5\sin 3\theta + 6\sqrt{3}\cos 3\theta \equiv R\cos\alpha \sin 3\theta + R\sin\alpha \cos 3\theta$$
$$\therefore 5 = R\cos\alpha \text{ and } 6\sqrt{3} = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{5}{R} \text{ and } \sin\alpha = \frac{6\sqrt{3}}{R}$$



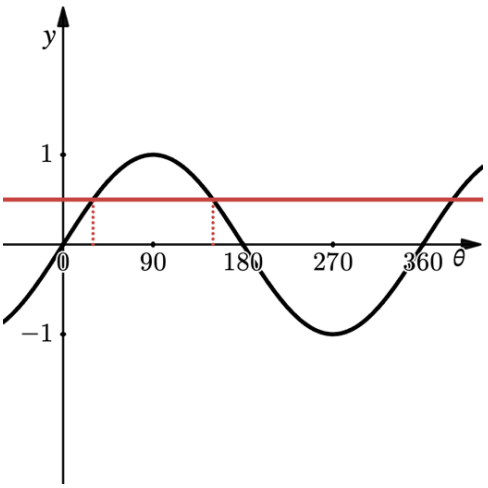
$$R = \sqrt{5^2 + (6\sqrt{3})^2} = \sqrt{133}$$
$$\alpha = \arctan\left(\frac{6\sqrt{3}}{5}\right) = 64.306\ldots^\circ$$
$$\therefore 5\sin 3\theta + 6\sqrt{3}\cos 3\theta = \sqrt{133}\sin(3\theta + 64.306\ldots^\circ)$$

$5\sin 3\theta + 6\sqrt{3}\cos 3\theta = 1$ for $0^\circ < \theta < 240^\circ$

$$\Rightarrow \sqrt{133}\sin(3\theta + 64.306\ldots^\circ) = 1$$
$$\Rightarrow \sin(3\theta + 64.306\ldots^\circ) = \frac{1}{\sqrt{133}}$$

Find first value between 0° and 360°:

$$3\theta + 64.306\ldots^\circ = \arcsin\frac{1}{\sqrt{133}} = 4.974\ldots^\circ$$



Find second value between 0° and 360°:

$$3\theta + 64.306\ldots^\circ = 180^\circ - 4.974\ldots^\circ = 175.025\ldots^\circ$$

Apply transformation:

$$3\theta + 64.306\ldots^\circ = 4.974\ldots^\circ \Rightarrow \theta = -19.777\ldots^\circ \quad \text{(not in range)}$$

$$3\theta + 64.306\ldots^\circ = 175.025\ldots^\circ \Rightarrow \theta = 36.906\ldots^\circ \quad \text{(in range)}$$

Add period:

$$\theta = -19.777\ldots^\circ + 120^\circ = 100.222\ldots^\circ \quad \text{(in range)}$$

$$\theta = 100.222\ldots^\circ + 120^\circ = 220.222\ldots^\circ \quad \text{(in range)}$$

$$\theta = 36.906\ldots^\circ + 120^\circ = 156.906\ldots^\circ \quad \text{(in range)}$$

$\theta = \{36.9^\circ, 100^\circ, 157^\circ, 220^\circ\}$ to 3sf.

Alternatively:

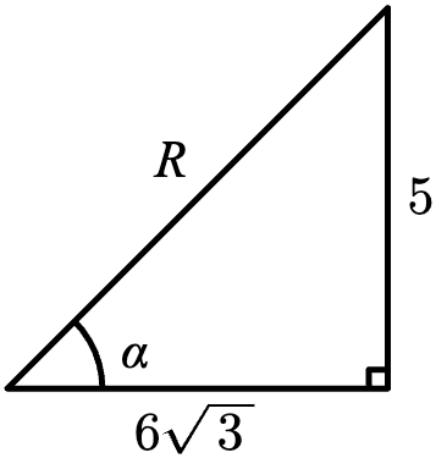
$$R\cos(3\theta - \alpha) \equiv R(\cos 3\theta \cos\alpha + \sin 3\theta \sin\alpha)$$

$$\equiv R\cos\alpha\cos 3\theta + R\sin\alpha\sin 3\theta$$

$$6\sqrt{3}\cos 3\theta + 5\sin 3\theta \equiv R\cos\alpha\cos 3\theta + R\sin\alpha\sin 3\theta$$

$$\therefore 6\sqrt{3} = R\cos\alpha \text{ and } 5 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{6\sqrt{3}}{R} \text{ and } \sin\alpha = \frac{5}{R}$$



$$R = \sqrt{(6\sqrt{3})^2 + 5^2} = \sqrt{133}$$

$$\alpha = \arctan\left(\frac{5}{6\sqrt{3}}\right) = 25.693\ldots^\circ$$

$$\therefore 6\sqrt{3}\cos 3\theta + 5\sin 3\theta = \sqrt{133}\cos(3\theta - 25.693\ldots^\circ)$$

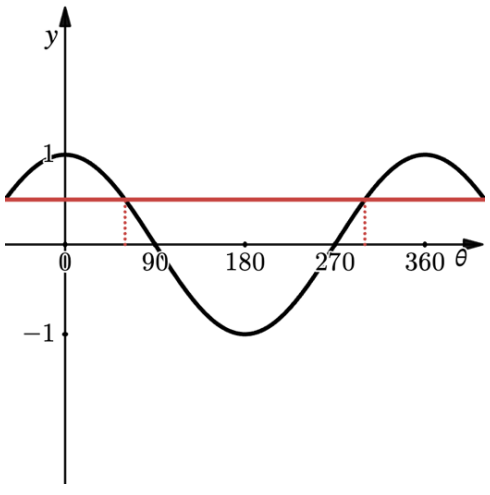
$6\sqrt{3}\cos 3\theta + 5\sin 3\theta = 1$ for $0^\circ < \theta < 240^\circ$

$$\Rightarrow \sqrt{133}\cos(3\theta - 25.693\ldots^\circ) = 1$$

$$\Rightarrow \cos(3\theta - 25.693\ldots^\circ) = \frac{1}{\sqrt{133}}$$

Find first value between 0° and 360°:

$$3\theta - 25.693\ldots^\circ = \arccos\frac{1}{\sqrt{133}} = 85.025\ldots^\circ$$



Find second value between 0° and 360°:

$$3\theta - 25.693\ldots^\circ = 360^\circ - 85.025\ldots^\circ = 274.974\ldots^\circ$$

Apply transformation:

$$3\theta - 25.693\ldots^\circ = 85.025\ldots^\circ \Rightarrow \theta = 36.906\ldots^\circ \quad \text{(in range)}$$

$$3\theta - 25.693\ldots^\circ = 274.974\ldots^\circ \Rightarrow \theta = 100.222\ldots^\circ \quad \text{(in range)}$$

Add period:

$$\theta = 36.906\ldots^\circ + 120^\circ = 156.906\ldots^\circ$$

$$\theta = 100.222\ldots^\circ + 120^\circ = 220.222\ldots^\circ$$

$\theta = \{36.9^\circ, 100^\circ, 157^\circ, 220^\circ\}$ to 3sf.

12 d)

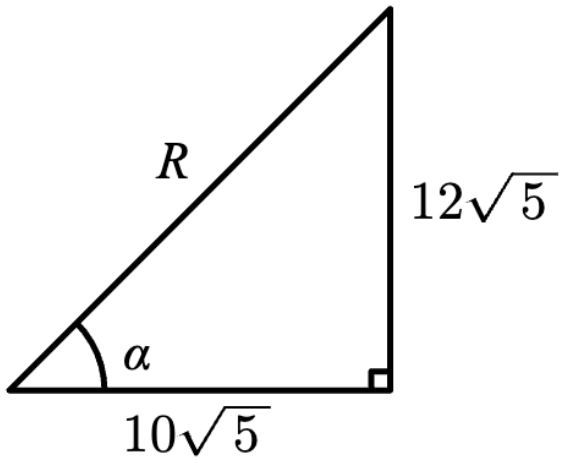
$$R\cos(2\theta + \alpha) \equiv R(\cos 2\theta \cos\alpha - \sin 2\theta \sin\alpha)$$

$$\equiv R\cos\alpha\cos 2\theta - R\sin\alpha\sin 2\theta$$

$$10\sqrt{5}\cos 2\theta - 12\sqrt{5}\sin 2\theta \equiv R\cos\alpha\cos 2\theta - R\sin\alpha\sin 2\theta$$

$$\therefore 10\sqrt{5} = R\cos\alpha \text{ and } 12\sqrt{5} = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{10\sqrt{5}}{R} \text{ and } \sin\alpha = \frac{12\sqrt{5}}{R}$$



$$R = \sqrt{(10\sqrt{5})^2 + (12\sqrt{5})^2} = 2\sqrt{305}$$

$$\alpha = \arctan\left(\frac{12\sqrt{5}}{10\sqrt{5}}\right) = 0.8760\ldots$$

$$\therefore 10\sqrt{5}\cos 2\theta - 12\sqrt{5}\sin 2\theta = 2\sqrt{305}\cos(2\theta + 0.8760\ldots)$$

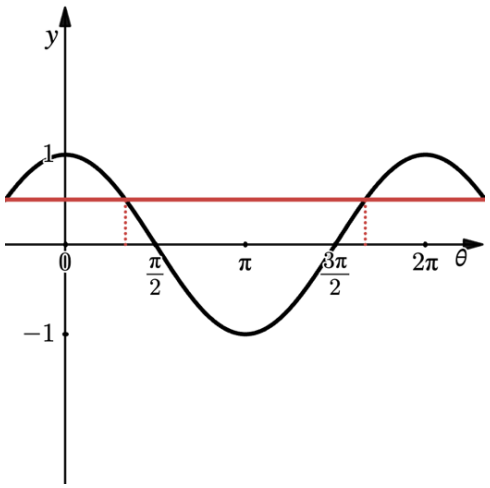
$10\sqrt{5}\cos 2\theta - 12\sqrt{5}\sin 2\theta = 5$ for $0 < \theta < 2\pi$

$$\Rightarrow 2\sqrt{305}\cos(2\theta + 0.8760\ldots) = 5$$

$$\Rightarrow \cos(2\theta + 0.8760\ldots) = \frac{5}{2\sqrt{305}}$$

Find first value between 0 and 2π:

$$2\theta + 0.8760\ldots = \arccos\frac{5}{2\sqrt{305}} = 1.4271\ldots$$



Find second value between 0 and 2π:

$$2\theta + 0.8760\ldots = 2\pi - 1.4271\ldots = 4.8560\ldots$$

Apply transformation:

$$2\theta + 0.8760\ldots = 1.4271\ldots \Rightarrow \theta = 0.2755\ldots \quad \text{(in range)}$$

$$2\theta + 0.8760\ldots = 4.8560\ldots \Rightarrow \theta = 1.9899\ldots \quad \text{(in range)}$$

Add period:

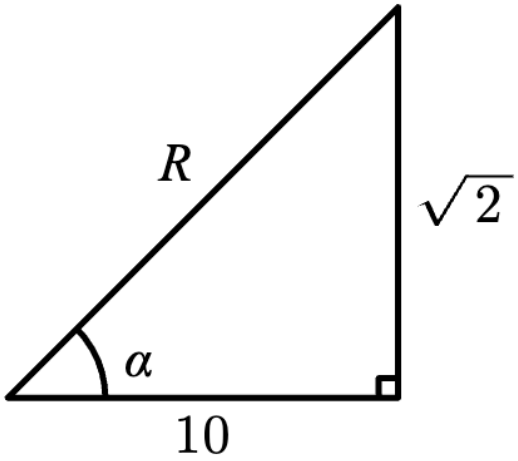
$$\theta = 0.2755\ldots + \pi = 3.4171\ldots \quad \text{(in range)}$$

$$\theta = 1.9899\ldots + \pi = 5.1315\ldots \quad \text{(in range)}$$

$\theta = \{0.276, 1.99, 3.42, 5.13\}$ to 3sf.

13)

$$R\sin(x-\alpha) \equiv R(\sin x\cos\alpha - \cos x\sin\alpha)$$
$$\equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$10\sin x - \sqrt{2}\cos x \equiv R\cos\alpha\sin x - R\sin\alpha\cos x$$
$$\therefore 10 = R\cos\alpha \text{ and } \sqrt{2} = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{10}{R} \text{ and } \sin\alpha = \frac{\sqrt{2}}{R}$$



$$R = \sqrt{10^2 + (\sqrt{2})^2} = \sqrt{102}$$
$$\alpha = \arctan\left(\frac{\sqrt{2}}{10}\right) = 0.140 \text{ to 3sf}$$
$$\therefore 10\sin x - \sqrt{2}\cos x = \sqrt{102}\sin(x - 0.140)$$
$$y = \sin x \mapsto y = 4 + \sqrt{102}\sin(x - 0.140)$$

Stretch, parallel to the y-axis, scale factor $\sqrt{102}$ followed by a Translation by the vector $\begin{bmatrix}0.140\\4\end{bmatrix}$.

14)

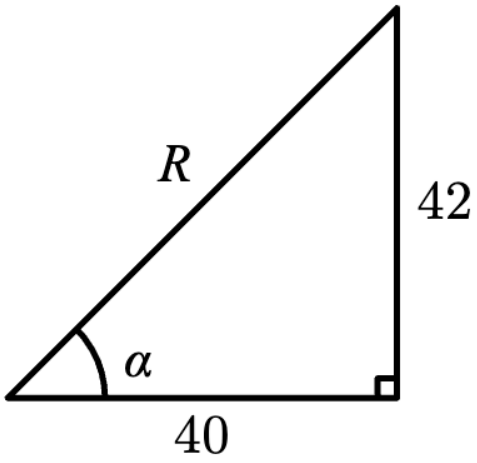
$$R\cos(5x+\alpha) \equiv R(\cos 5x\cos\alpha - \sin 5x\sin\alpha)$$

$$\equiv R\cos\alpha\cos 5x - R\sin\alpha\sin 5x$$

$$40\cos 5x - 42\sin 5x \equiv R\cos\alpha\cos 5x - R\sin\alpha\sin 5x$$

$$\therefore 40 = R\cos\alpha \text{ and } 42 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{40}{R} \text{ and } \sin\alpha = \frac{42}{R}$$



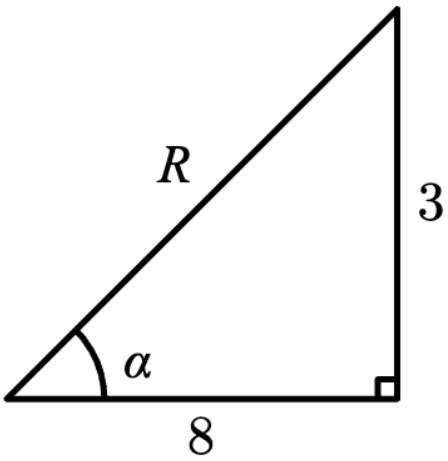
$$R = \sqrt{40^2 + 42^2} = 58$$

$$\alpha = \arctan\left(\frac{42}{40}\right) = 0.810 \text{ to 3sf}$$

$$\therefore 40\cos 5x - 42\sin 5x = 58\cos(5x + 0.810)$$

15)

$$R\cos(\frac{1}{2}x-\alpha) \equiv R(\cos\frac{1}{2}x\cos\alpha+\sin\frac{1}{2}x\sin\alpha)$$
$$\equiv R\cos\alpha\cos\frac{1}{2}x+R\sin\alpha\sin\frac{1}{2}x$$
$$8\cos\frac{1}{2}x+3\sin\frac{1}{2}x \equiv R\cos\alpha\cos\frac{1}{2}x+R\sin\alpha\sin\frac{1}{2}x$$
$$\therefore 8=R\cos\alpha \text{ and } 3=R\sin\alpha$$
$$\Rightarrow \cos\alpha=\frac{8}{R} \text{ and } \sin\alpha=\frac{3}{R}$$



$$R=\sqrt{8^2+3^2}=\sqrt{73}$$
$$\alpha=\arctan\left(\frac{3}{8}\right)=0.359 \text{ to 3sf}$$
$$\therefore 8\cos\frac{1}{2}x+3\sin\frac{1}{2}x=\sqrt{73}\cos\left(\frac{1}{2}x-0.359\right)$$

16)

$$\sqrt{p^2+q^2}=4 \Rightarrow p^2+q^2=16$$
$$\sqrt{2}-\sqrt{6}=p\sin\left(-\frac{\pi}{4}\right)+q\cos\left(-\frac{\pi}{4}\right)$$
$$\Rightarrow \sqrt{2}-\sqrt{6}=-\frac{\sqrt{2}}{2}p+\frac{\sqrt{2}}{2}q$$
$$\Rightarrow \frac{\sqrt{2}}{2}q=\sqrt{2}-\sqrt{6}+\frac{\sqrt{2}}{2}p$$
$$\Rightarrow q=2-2\sqrt{3}+p$$

Substitute:

$$p^2+(2-2\sqrt{3}+p)^2=16$$
$$\Rightarrow p^2+16-8\sqrt{3}+2(2-2\sqrt{3})p+p^2=16$$
$$\Rightarrow 2p^2+2(2-2\sqrt{3})p-8\sqrt{3}=0$$
$$\Rightarrow p^2+(2-2\sqrt{3})p-4\sqrt{3}=0$$

$$\Rightarrow (p+2)(p-2\sqrt{3}) = 0 \Rightarrow p = -2 \text{ or } p = 2\sqrt{3}$$

$p > 0$ so $p = 2\sqrt{3}$.
When $p = 2\sqrt{3}$, then $q = 2 - 2\sqrt{3} + 2\sqrt{3} = 2$.

$$\therefore y = 2\sqrt{3}\sin x + 2\cos x$$

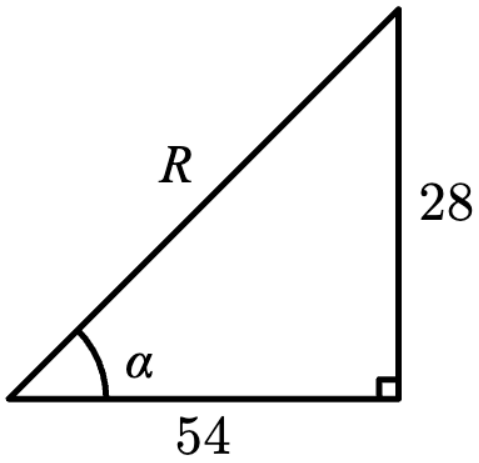
17 a)

$$h = k + 28\cos 4t + 54\sin 4t$$

When $t = 0, h = 117$: $117 = k + 28 \Rightarrow k = 89$.

17 b)

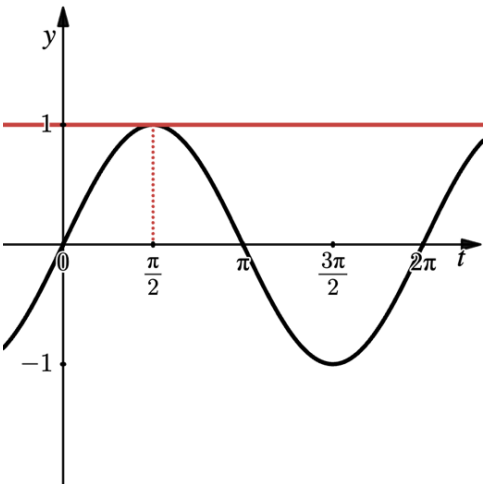
$$R\sin(4t+\alpha) \equiv R(\sin 4t\cos\alpha + \cos 4t\sin\alpha)$$
$$\equiv R\cos\alpha\sin 4t + R\sin\alpha\cos 4t$$
$$54\sin 4t + 28\cos 4t \equiv R\cos\alpha\sin 4t + R\sin\alpha\cos 4t$$
$$\therefore 54 = R\cos\alpha \text{ and } 28 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{54}{R} \text{ and } \sin\alpha = \frac{28}{R}$$



$$R = \sqrt{54^2 + 28^2} = 10\sqrt{37}$$
$$\alpha = \arctan\left(\frac{28}{54}\right) = 0.4783\ldots$$
$$\therefore 54\sin 4t + 28\cos 4t = 10\sqrt{37}\sin(4t + 0.4783\ldots)$$

The maximum height is $89 + 10\sqrt{37}$ metres.
Solve $89 + 10\sqrt{37}\sin(4t + 0.4783\ldots) = 89 + 10\sqrt{37}$

$$\Rightarrow \sin(4t + 0.4783\ldots) = 1$$



$$4t + 0.4783\ldots = \frac{\pi}{2}$$

Apply transformation:

$$t = 0.2731\ldots = 0.273 \text{ seconds to 3sf}$$

Alternatively:

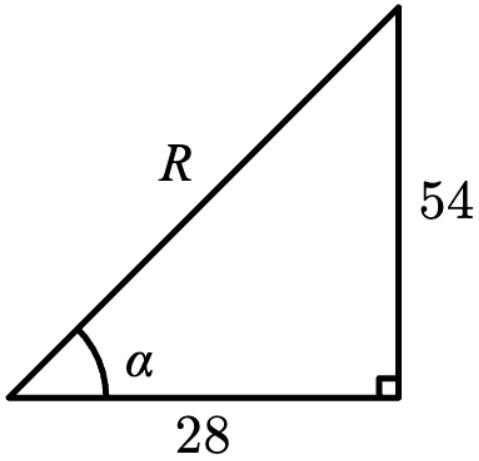
$$R\cos(4t-\alpha) \equiv R(\cos 4t\cos\alpha + \sin 4t\sin\alpha)$$

$$\equiv R\cos\alpha\cos 4t + R\sin\alpha\sin 4t$$

$$28\cos 4t + 54\sin 4t \equiv R\cos\alpha\cos 4t + R\sin\alpha\sin 4t$$

$$\therefore 28 = R\cos\alpha \text{ and } 54 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{28}{R} \text{ and } \sin\alpha = \frac{54}{R}$$



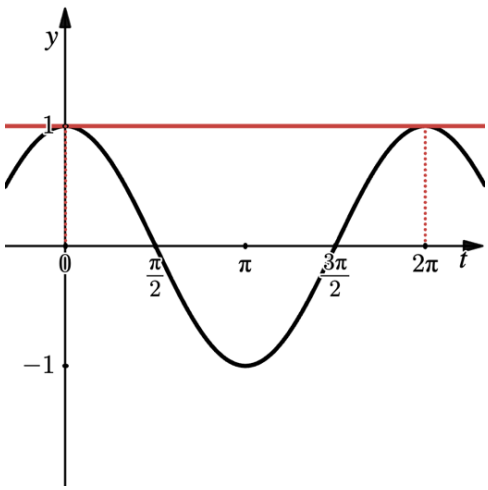
$$R = \sqrt{28^2 + 54^2} = 10\sqrt{37}$$

$$\alpha = \arctan\left(\frac{54}{28}\right) = 1.0924\ldots$$

$$\therefore 28\cos 4t + 54\sin 4t = 10\sqrt{37}\cos(4t - 1.0924\ldots)$$

The maximum height is $89 + 10\sqrt{37}$ metres.
Solve $89 + 10\sqrt{37}\cos(4t - 1.0924\ldots) = 89 + 10\sqrt{37}$

$$\Rightarrow \cos(4t - 1.0924\ldots) = 1$$



$$4t - 1.0924\ldots = 0$$

Apply transformation:

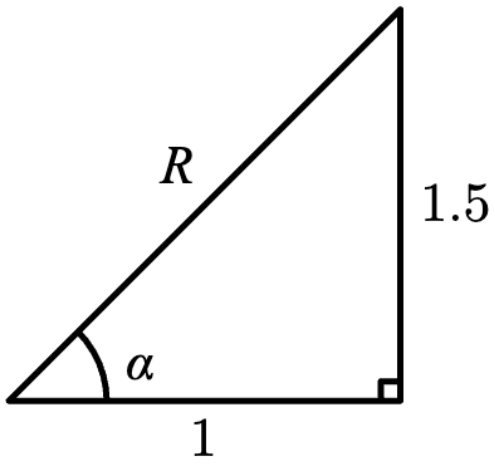
$$t = 0.2731\ldots = 0.273 \text{ seconds to 3sf}$$

18 a)

Minimum Depth is $4 - \sqrt{1^2 + 1.5^2} = (4 - \frac{\sqrt{13}}{2})$ metres.

18 b)

$$R\sin(\tfrac{\pi}{6}t - 2 + \alpha) \equiv R\left(\sin(\tfrac{\pi}{6}t - 2)\cos\alpha + \cos(\tfrac{\pi}{6}t - 2)\sin\alpha\right)$$
$$\equiv R\cos\alpha\sin(\tfrac{\pi}{6}t - 2) + R\sin\alpha\cos(\tfrac{\pi}{6}t - 2)$$
$$\sin(\tfrac{\pi}{6}t - 2) + 1.5\cos(\tfrac{\pi}{6}t - 2) \equiv R\cos\alpha\sin(\tfrac{\pi}{6}t - 2) + R\sin\alpha\cos(\tfrac{\pi}{6}t - 2)$$
$$\therefore 1 = R\cos\alpha \text{ and } 1.5 = R\sin\alpha$$
$$\Rightarrow \cos\alpha = \frac{1}{R} \text{ and } \sin\alpha = \frac{1.5}{R}$$

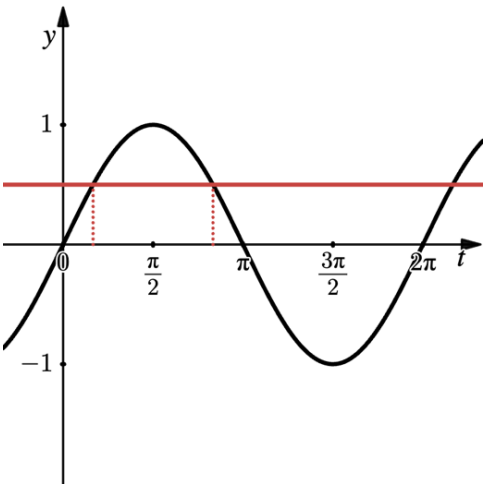


$$R = \sqrt{1^2 + 1.5^2} = \frac{\sqrt{13}}{2}$$
$$\alpha = \arctan\left(\frac{1.5}{1}\right) = 0.9827\ldots$$
$$\therefore \sin(\tfrac{\pi}{6}t - 2) + 1.5\cos(\tfrac{\pi}{6}t - 2) \equiv \frac{\sqrt{13}}{2}\sin(\tfrac{\pi}{6}t - 2 + 0.9827\ldots)$$
$$\equiv \frac{\sqrt{13}}{2}\sin(\tfrac{\pi}{6}t - 1.0172\ldots)$$

Solve $4 + \frac{\sqrt{13}}{2}\sin(\frac{\pi}{6}t - 1.0172\ldots) = 5$

$$\Rightarrow \sin(\tfrac{\pi}{6}t - 1.0172\ldots) = \frac{2}{\sqrt{13}}$$

$$\tfrac{\pi}{6}t - 1.0172\ldots = \arcsin\frac{2}{\sqrt{13}} = 0.5880\ldots$$



Apply transformation:

$$\tfrac{\pi}{6}t - 1.0172\ldots = 0.5880\ldots \Rightarrow t = 3.0657\ldots$$

This represents 03:04 am to the nearest minute.
Add period:

$$t = 3.0657\ldots + 12 = 15.0657\ldots$$

This represents 03:04 pm to the nearest minute.
So Derek must wait 2 hours and 4 minutes, to the nearest minute.
Alternatively:

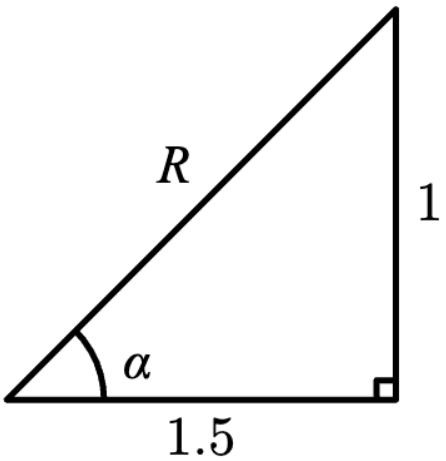
$$R\cos(\tfrac{\pi}{6}t - 2 + \alpha) \equiv R\left(\cos(\tfrac{\pi}{6}t - 2)\cos\alpha + \sin(\tfrac{\pi}{6}t - 2)\sin\alpha\right)$$

$$\equiv R\cos\alpha\cos(\tfrac{\pi}{6}t - 2) + R\sin\alpha\sin(\tfrac{\pi}{6}t - 2)$$

$$1.5\cos(\tfrac{\pi}{6}t - 2) + \sin(\tfrac{\pi}{6}t - 2) \equiv R\cos\alpha\cos(\tfrac{\pi}{6}t - 2) + R\sin\alpha\sin(\tfrac{\pi}{6}t - 2)$$

$$\therefore 1.5 = R\cos\alpha \text{ and } 1 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{1.5}{R} \text{ and } \sin\alpha = \frac{1}{R}$$



$$R = \sqrt{1.5^2 + 1^2} = \frac{\sqrt{13}}{2}$$

$$\alpha = \arctan\left(\frac{1}{1.5}\right) = 0.5880\ldots$$

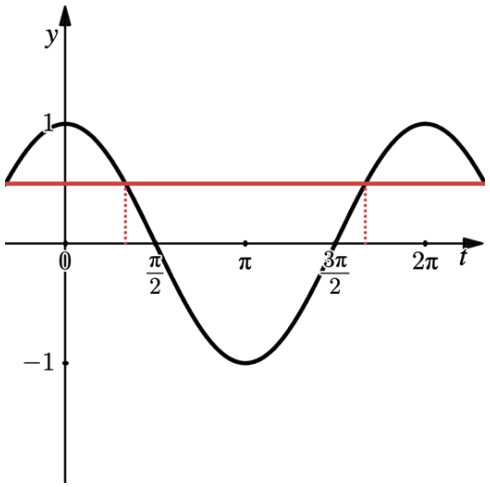
$$\therefore 1.5\cos(\tfrac{\pi}{6}t-2)+\sin(\tfrac{\pi}{6}t-2) \equiv \frac{\sqrt{13}}{2}\cos(\tfrac{\pi}{6}t-2-0.5880\ldots)$$
$$\equiv \frac{\sqrt{13}}{2}\cos(\tfrac{\pi}{6}t-2.5880\ldots)$$

Solve $4+\frac{\sqrt{13}}{2}\cos(\frac{\pi}{6}t-2.5880\ldots)=5$

$$\Rightarrow \cos(\tfrac{\pi}{6}t-2.5880\ldots)=\frac{2}{\sqrt{13}}$$

Find first value between 0 and 2π:

$$\tfrac{\pi}{6}t-2.5880\ldots=\arccos\frac{2}{\sqrt{13}}=0.9827\ldots$$



Find second value between 0 and 2π:

$$\tfrac{\pi}{6}t-2.5880\ldots=2\pi-0.9827\ldots=5.3003\ldots$$

Apply transformation:

$$\tfrac{\pi}{6}t-2.5880\ldots=0.9827\ldots\Rightarrow t=6.8197\ldots$$

Note that this represents 06:49 am to the nearest minute and this is when the ferry stops running.

$$\tfrac{\pi}{6}t-2.5880\ldots=5.3003\ldots\Rightarrow t=15.0657\ldots$$

This represents 03:04 pm to the nearest minute and this is the 2nd time during the day that the ferry can start work.
So Derek must wait 2 hours and 4 minutes, to the nearest minute.

19 a)

Maximum displacement $=50+\sqrt{40^2+30^2}=100$ mm.

19 b)

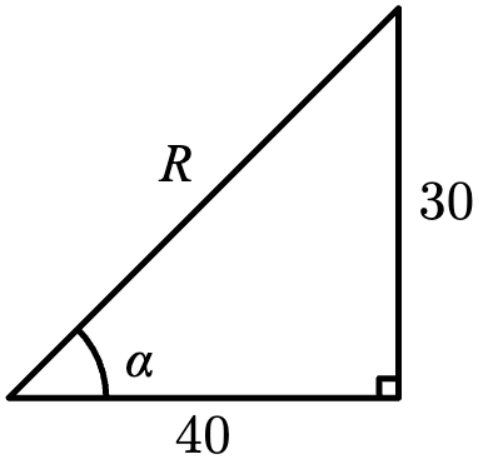
$$R\cos(20t+\alpha) \equiv R(\cos 20t\cos\alpha - \sin 20t\sin\alpha)$$

$$\equiv R\cos\alpha\cos 20t - R\sin\alpha\sin 20t$$

$$40\cos 20t - 30\sin 20t \equiv R\cos\alpha\cos 20t - R\sin\alpha\sin 20t$$

$$\therefore 40 = R\cos\alpha \text{ and } 30 = R\sin\alpha$$

$$\Rightarrow \cos\alpha = \frac{40}{R} \text{ and } \sin\alpha = \frac{30}{R}$$



$$R = \sqrt{40^2 + 30^2} = 50$$

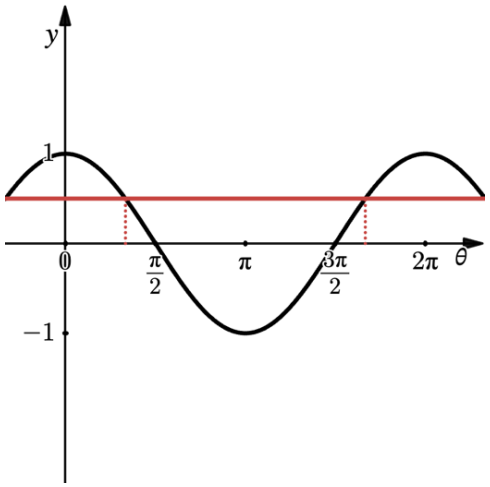
$$\alpha = \arctan\left(\frac{30}{40}\right) = 0.6435\ldots$$

$$\therefore 40\cos 20t - 30\sin 20t = 50\cos(20t + 0.6435\ldots)$$

Solve $50 + 50\cos(20t + 0.6435\ldots) = 80$

$$\Rightarrow \cos(20t + 0.6435\ldots) = \frac{3}{5}$$

$$20t + 0.6435\ldots = \arccos\frac{3}{5} = 0.9272\ldots$$



Also consider

$$20t + 0.6435\ldots = -0.9272\ldots$$

Apply transformation:

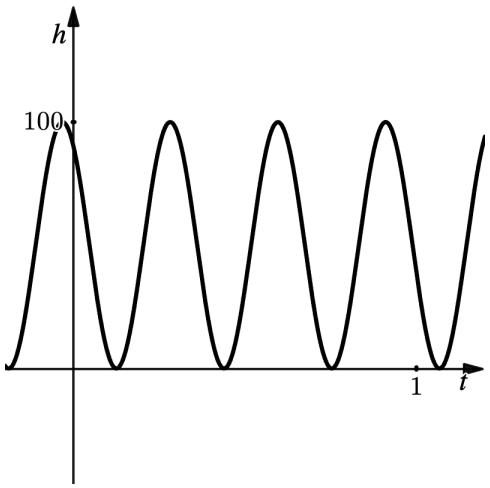
$$20t + 0.6435\ldots = 0.9272\ldots \Rightarrow \theta = 0.0141\ldots$$

$$20t + 0.6435\ldots = -0.9272\ldots \Rightarrow \theta = -0.0785\ldots$$

The time that A is more than 80 mm above its lowest position is $T = 0.0141\ldots + 0.0785\ldots = 0.0927\ldots = 0.0927$ seconds to 3sf.

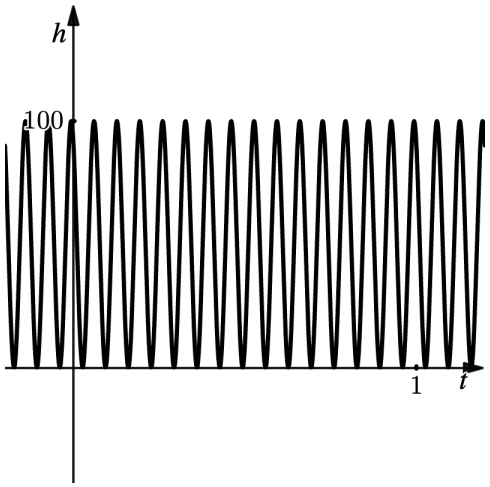
19 c)

Currently, the model has the piston making $\frac{20}{2\pi} \approx 3.183$ cycles per second.



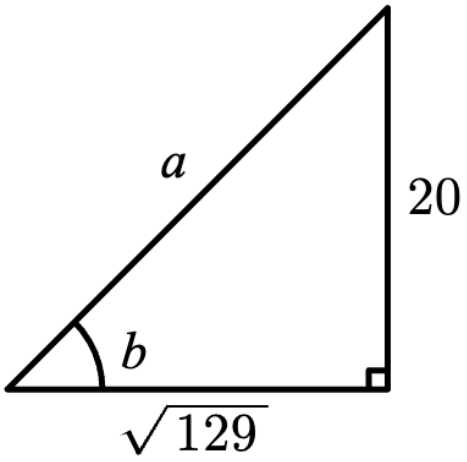
15 cycles per second corresponds to replacing $20t$ with $15 \times 2\pi t = 30\pi t$:

$$h = 50 + 40\cos(30\pi t) - 30\sin(30\pi t)$$



20 a)

$$a\cos(\theta + b) \equiv a(\cos\theta\cos b - \sin\theta\sin b)$$
$$\equiv a\cos b\cos\theta - a\sin b\sin\theta$$
$$\sqrt{129}\cos\theta - 20\sin\theta \equiv a\cos b\cos\theta - a\sin b\sin\theta$$
$$\therefore \sqrt{129} = a\cos b \text{ and } 20 = a\sin b$$
$$\Rightarrow \cos b = \frac{\sqrt{129}}{a} \text{ and } \sin b = \frac{20}{a}$$



$$a = \sqrt{(\sqrt{129})^2 + 20^2} = 23$$
$$b = \arctan\left(\frac{20}{\sqrt{129}}\right) = 60.40\ldots^\circ$$
$$\therefore \sqrt{129}\cos\theta - 20\sin\theta \equiv 23\cos(\theta + 60.40\ldots^\circ)$$

20 b)

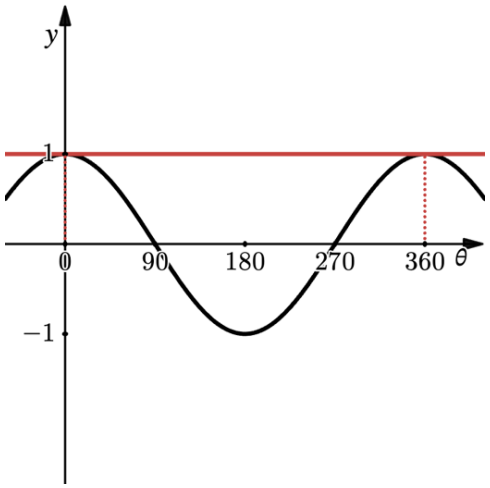
$$\therefore T = \sqrt{129}\cos(12t)^\circ - 20\sin(12t)^\circ + C$$
$$\Rightarrow T = 23\cos(12t + 60.40\ldots^\circ) + C$$
$$25 = 23 \times 1 + C \Rightarrow C = 2$$

20 c)

Maximum Humidity $= 10 \times 1 + 72 = 82$

$$82 = 10\cos(30t + 45°) + 72$$

$$\Rightarrow \cos(30t + 45°) = 1$$



$$30t + 45° = 0°$$

Apply transformation:

$$t = -1.5°$$

Add period:

$$t = -1.5° + 12° = 10.5°$$

When $t = 10.5°$,

$$T = \sqrt{129}\cos(30 \times 10.5)° - 20\sin(30 \times 10.5)° + 2 = 24.17\ldots° \text{ C} = 24.2°\text{C to 3sf}$$